Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
Lecture: CPR 47

Title: Serum Proteins and Disorders

Dr. Shellon Thomas
sthoma10@sgu.edu
St Georges University

Acknowledgements:
DR. MONIKA WINTER
monika.winter@northumbria.ac.uk
Dr. Sharmila Upadhya (shupadhya@sgu.edu)

Lecture: Serum Proteins and Disorders

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0318	Classify serum proteins based on electrophoretic mobility. Distinguish the functions of proteins that are found in the alpha1-globulin fraction (alpha1-antitrypsin, alpha-fetoprotein, retinol binding protein, transcortin).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0319	Predict three common causes of hypoalbuminemia (liver disease, nephrotic syndrome, protein malnutrition) and explain the biochemical basis for the occurrence of edema in hypoalbuminemia.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0320	Explain the pathophysiology of inherited alpha1-antitrypsin deficiency on the liver (gain-of-function) and lungs (loss of function).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0321	Distinguish the functions of proteins in the alpha 2-globulin fraction (alpha 2 macroglobulin, haptoglobin, ceruloplasmin). Describe serum protein changes in nephrotic syndrome and Wilson disease.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0322	Distinguish the functions of proteins found in the beta-globulin fraction (transferrin, hemopexin, beta lipoprotein).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0323	Enumerate the functions of proteins found in the gamma-globulin fraction (Immunoglobulins: IgG, IgM, IgA, IgD, IgE).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0324	Identify serum proteins that are increased following an inflammation. Give examples of positive and negative acute phase proteins.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0325	Interpret the utility of C-reactive protein as an inflammatory marker. Interpret the utility of haptoglobin levels as indicators of hemolysis.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0326	Recognize serum protein electrophoretic pattern in the following disorders: a. Acute and chronic inflammation (polyclonal gammopathy) b. Liver cirrhosis c. Monoclonal gammopathy (multiple myeloma) d. Nephrotic syndrome e. Hypogammaglobulinemia f. Alpha1- antitrypsin deficiency

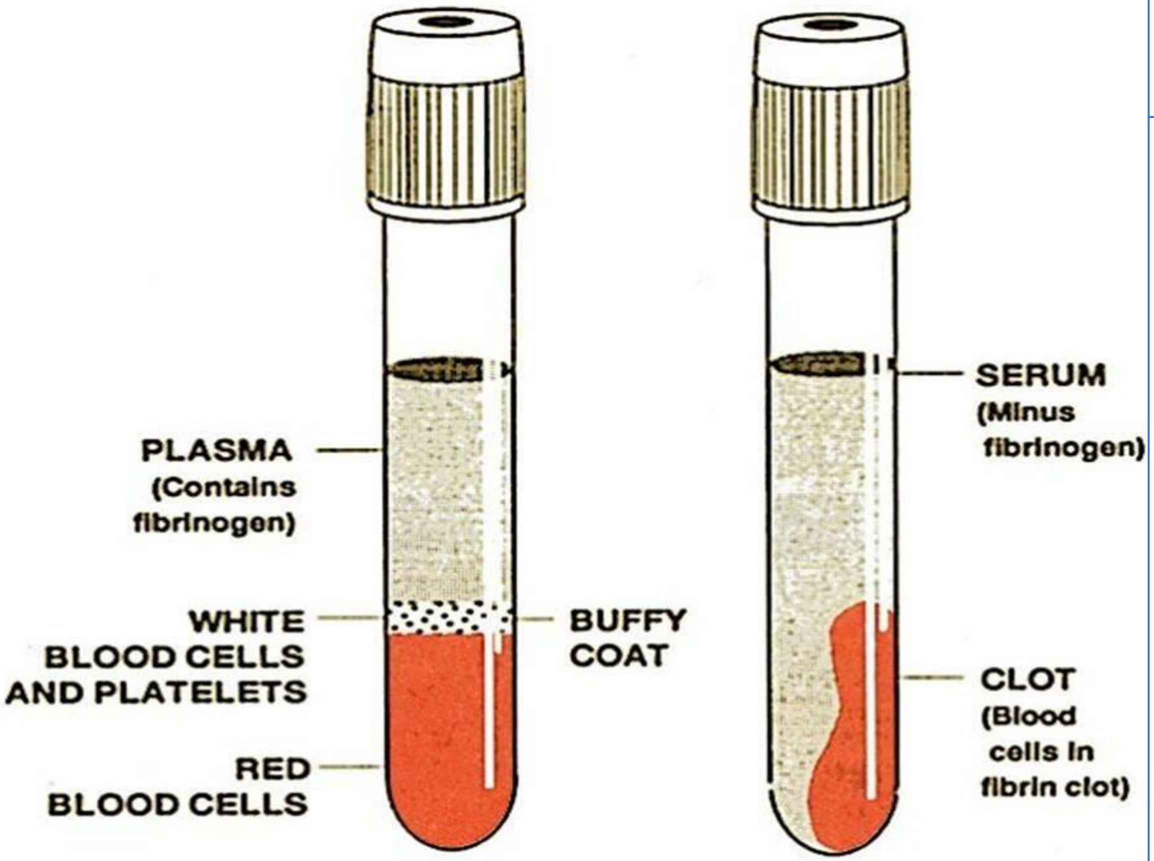
Recommended reading

- Blood and Plasma Proteins - Medical Biochemistry - ClinicalKey Student; Section: Plasma proteins
- https://vestnik.szd.si/index.php/ZdravVest/article/view/2899/3020#figures
- https://medlineplus.gov/genetics/condition/alpha-1-antitrypsin-deficiency/
- Alpha-1 Antitrypsin Deficiency - causes, symptoms, diagnosis, treatment, pathology – YouTube
- A Closer Look at Alpha-1 Antitrypsin Deficiency: Peggy's Story - YouTube

Difference between plasma and serum

Serum = Plasma – Clotting factors

PLASMA
• Anticoagulant + blood sample • Clear fluid after centrifugation • **Contains all clotting factors** • Anticoagulants: EDTA, citrate, heparin

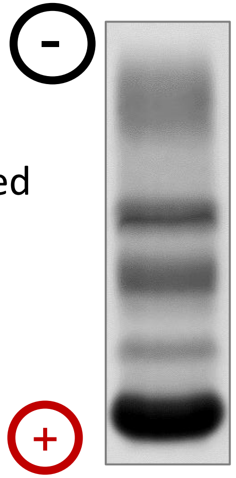


SERUM
• Blood sample (NO anticoagulant) • Clear fluid after centrifugation • **Contains NO clotting factors**

Serum Protein Electrophoresis (SPEP)

Separates serum proteins using electrical and endosmotic forces

Non-denaturing SPEP, pH 8.6



More negatively charged proteins **(Albumin)** will migrate furthest

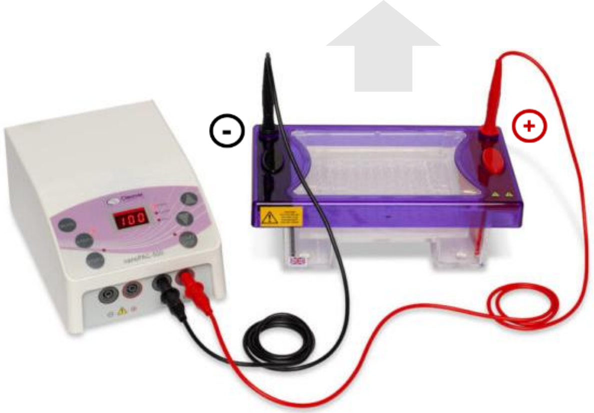
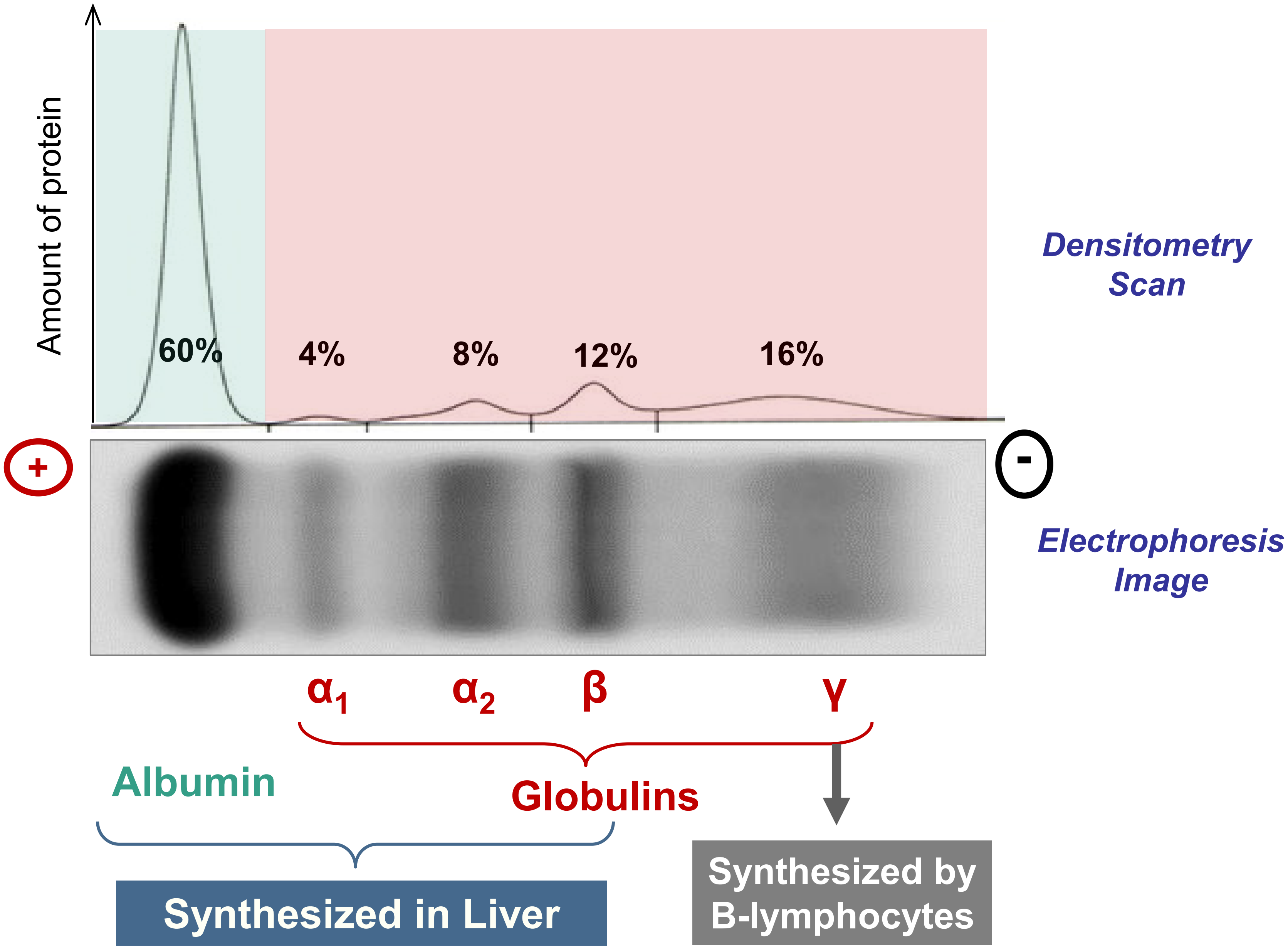


Image: https://clsresource.com/index.php/serum-proteins/

- At pH 8.6, serum proteins have net negative (-) charge
- Serum proteins move towards anode (+) depending on net charge
- **Albumin** moves fastest towards Anode
- Anode: Positive and attracts negatively charged anions;
- Cathode: Negative and attracts positively charged cations

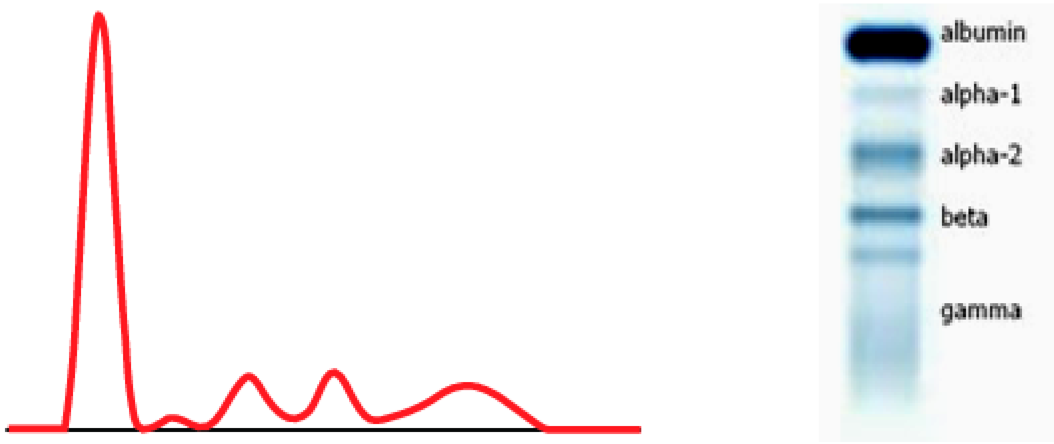
Classification of serum proteins: Two major groups - Albumin and Globulins



Serum protein electrophoresis

- **Component proteins in each fraction**
- **Function of each protein**
- **Clinical significance of each protein**
- **Serum protein electrophoresis patterns in disorders**
- Normal findings:
  - Total serum protein: 6-8 g/dL
  - Serum Albumin: 3.5-5 g/dL
  - Urine protein: Absent

Protein EP (serum)



Fraction	Result (g/dL)	Percentage (%)	Normal	Unit
Albumin	4.4	60.08	3.3–4.8	g/dL
Alpha1	0.1	1.46	0.1–0.3	g/dL
Alpha2	0.7	9.86	0.7–1.1	g/dL
Beta	0.8	10.39	0.8–1.2	g/dL
Gamma	1.3	18.21	0.7–1.8	g/dL
A/G ratio	1.51	1.51		
Total SPE	7.3	100.00	6.3–8.3	g/dL

Functions of plasma proteins

Maintenance of colloidal osmotic pressure (Oncotic pressure)

Albumin

Transport of ions, hormones, bilirubin, lipids

Albumin, α and β globulins

Inflammatory response and immunity

γ-globulins, Acute phase proteins

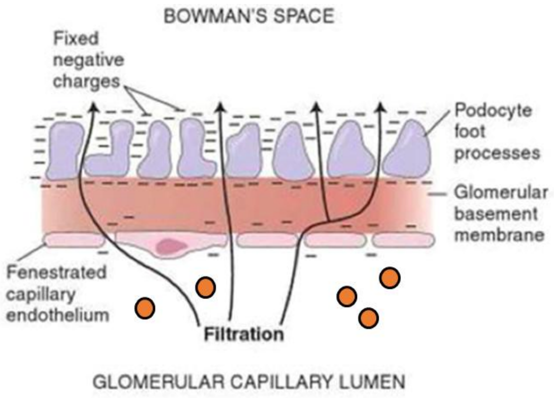
Hemostasis

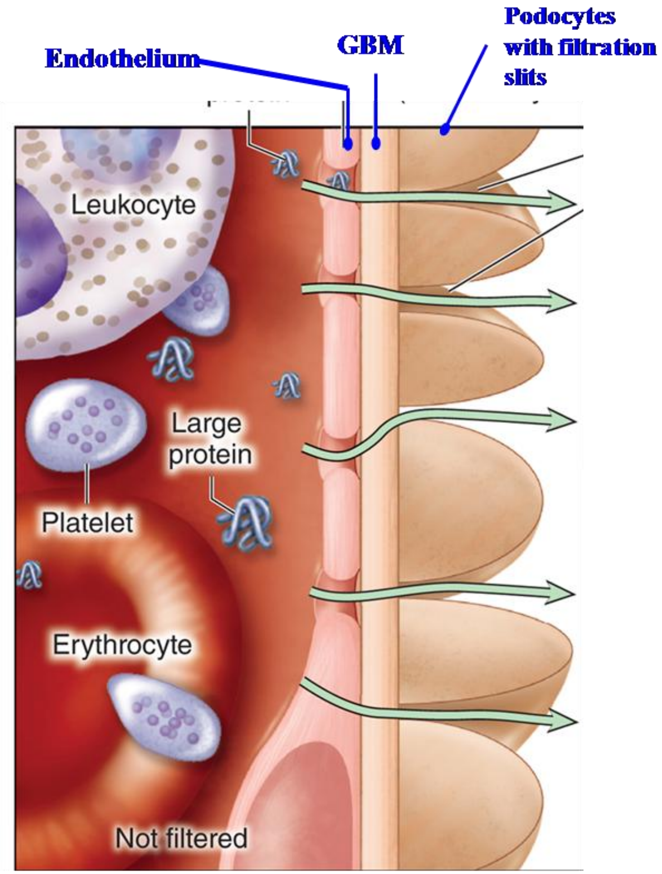
Clotting factors: prothrombin, fibrinogen

Functions of SERUM proteins

Albumin

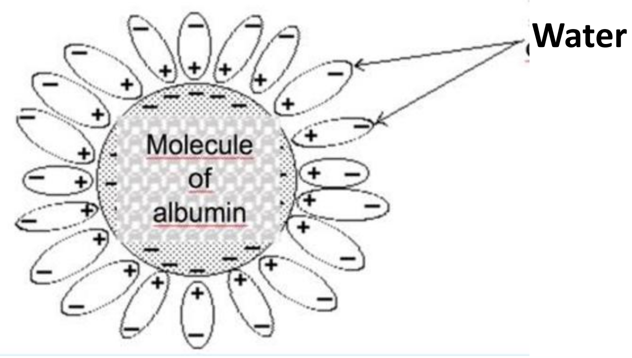
- About 55-60% of total serum protein
- Two major **functions**
  - Maintenance of colloidal osmotic pressure (oncotic pressure)
  - Carrier or transport protein
- **Albumin synthesized by liver**
- **Serum albumin level:**
  - **Indicator of nutritional status**
  - **Liver synthetic function**
  - **Indicator of glomerular basement membrane integrity**
- Negatively charged at pH 7.4
- Albumin NOT normally found in urine (Why?)
- **Albuminuria**: Glomerular basement membrane damage (**Nephrotic syndrome**; kidney diseases)



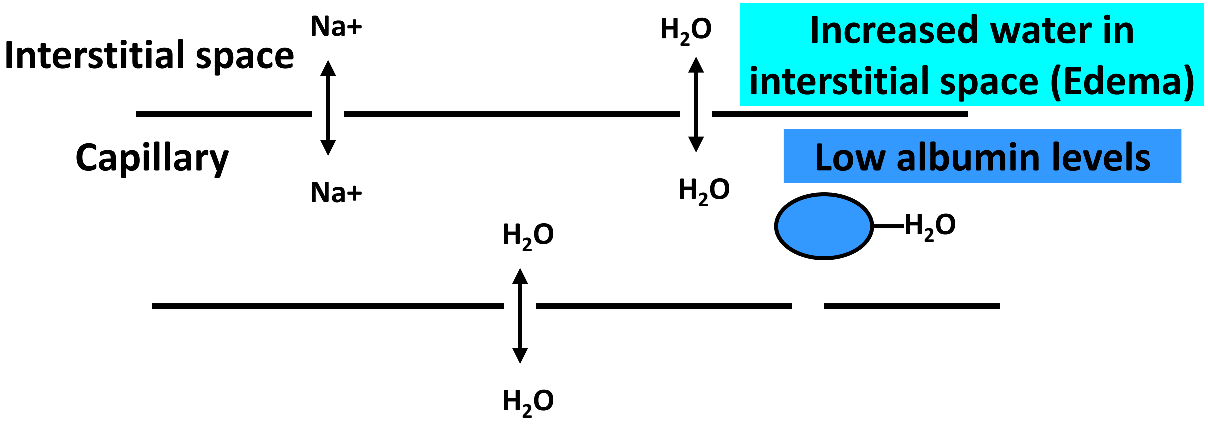


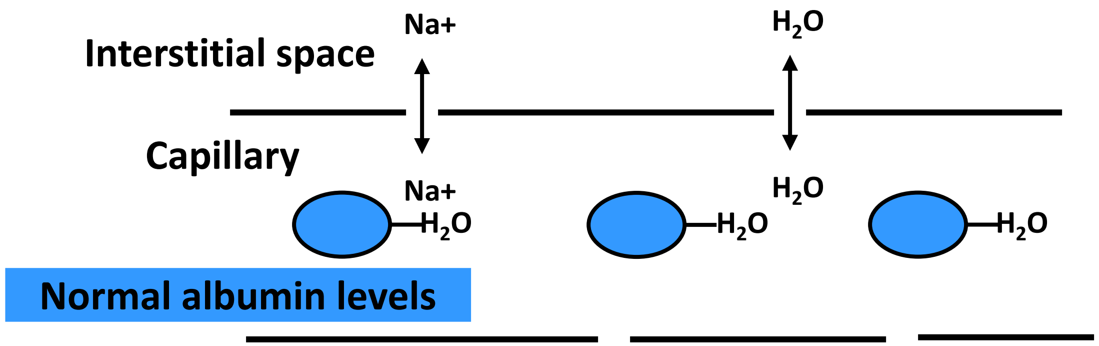
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0318; 0319

Albumin: Maintains colloidal osmotic pressure/ Oncotic pressure



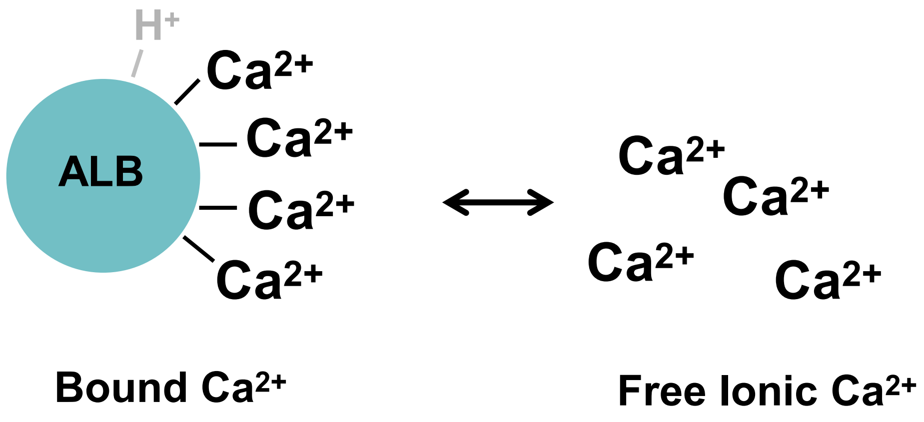
- Binds to water within capillaries
- Contributes to colloidal osmotic pressure (Oncotic pressure)
- **Hypoalbuminemia** → Low colloidal osmotic pressure (Oncotic pressure) → Fluid in interstitial space (Pitting edema)



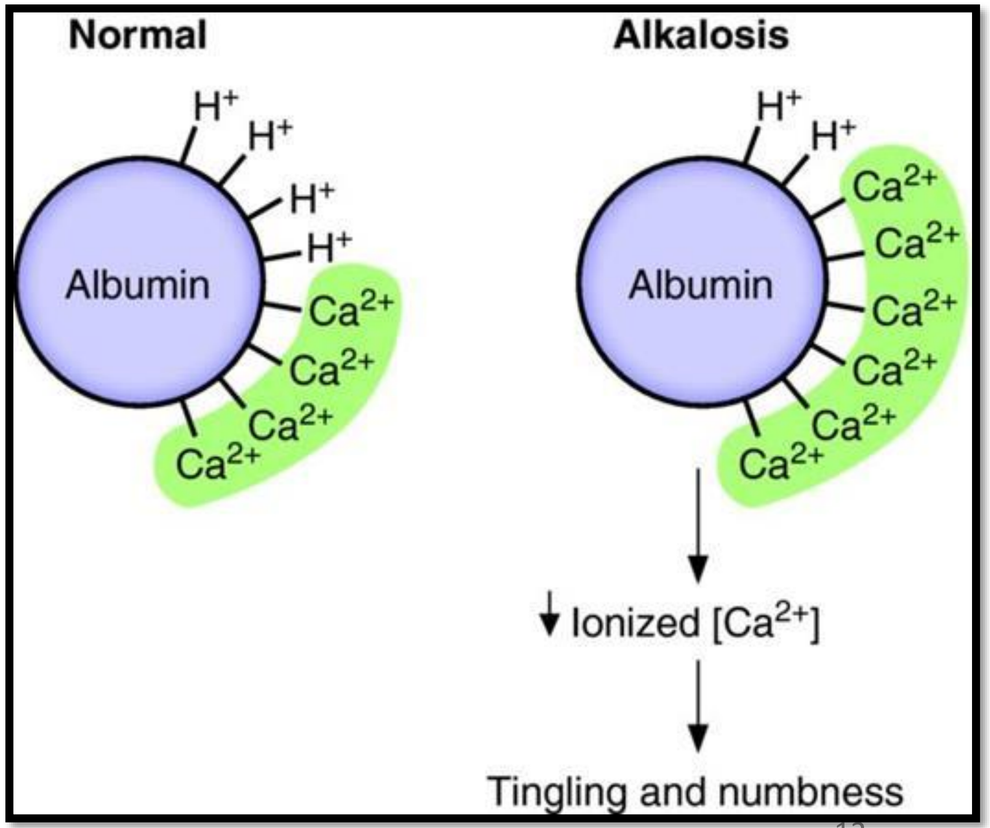


Pitting edema

Albumin: Transport protein



- Negative charges: binds to Ca^{2+}, reservoir of free/ ionic calcium (biologically active form)
- Ca^{2+} binding to albumin influenced by plasma pH;
  - Higher plasma pH (alkalosis), more negative charges on albumin and higher binding of Ca^{2+};
  - Alkalosis predisposes to **hypocalcemia and tetany** (**Lowers free/ ionic Ca^{2+}; Total calcium normal**)

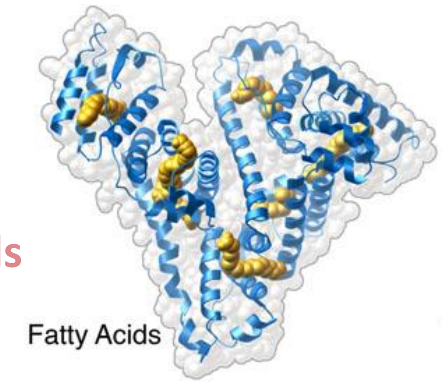


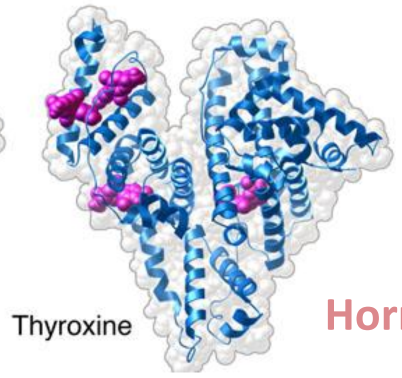
Albumin as a transport protein

- Binds lipid-soluble hormones (steroid and thyroid hormone), bilirubin, drugs, fatty acids
  - Weak interactions and can be displaced

Lipophilic compounds bound to hydrophobic clefts

Long chain fatty acids

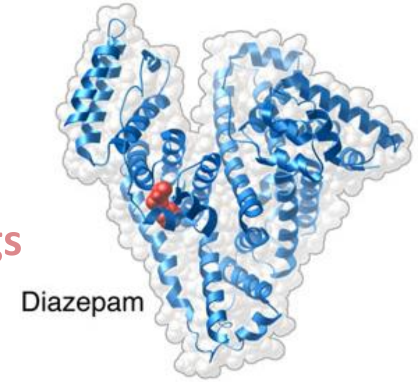


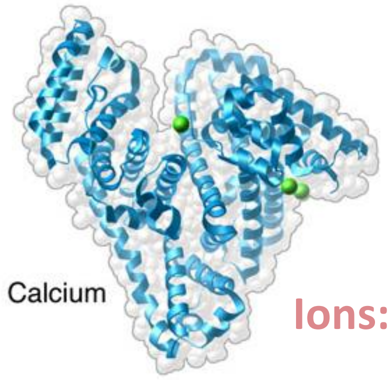


Hormones

Because of its negative charge albumin can bind and transport

Positively charged drugs





Ions: Calcium

Hypoalbuminemia

Decreased albumin synthesis

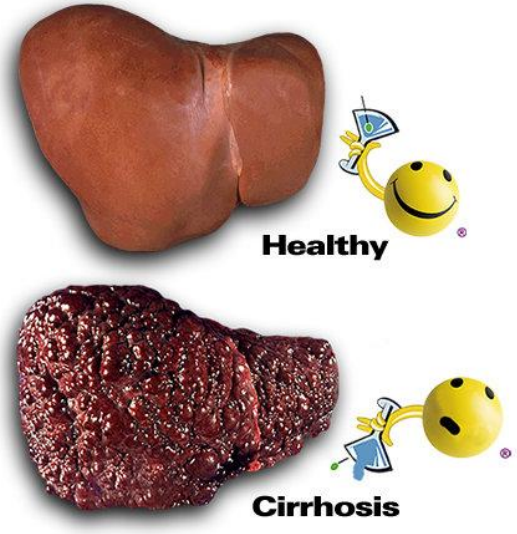
Increased albumin loss

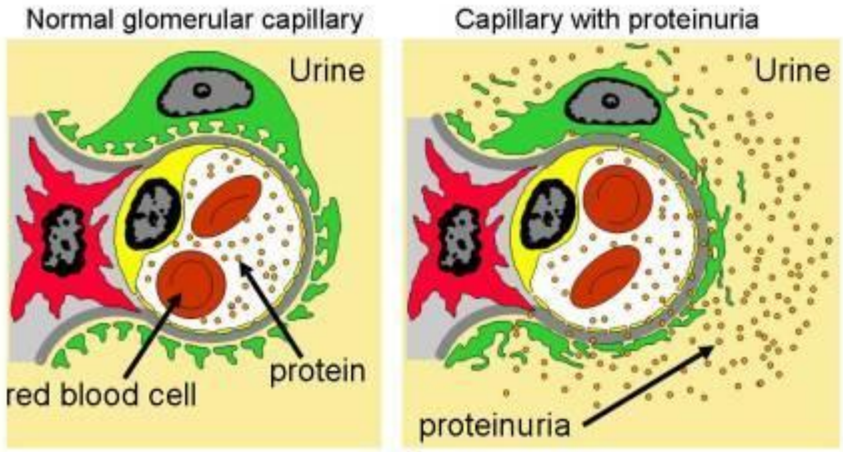
Low protein diet (Kwashiorkor)

Chronic liver disease (cirrhosis)

Severe burns (Increased vascular permeability in burnt areas)

Loss of albumin in urine **(Nephrotic syndrome**; kidney disease)





SOM.MK.I.BPM1.1.CPR.2.BCHM. 0318; 0319

Component Proteins in α, β and γ globulin fractions

Anode ⊕

Cathode ⊖

Albumin

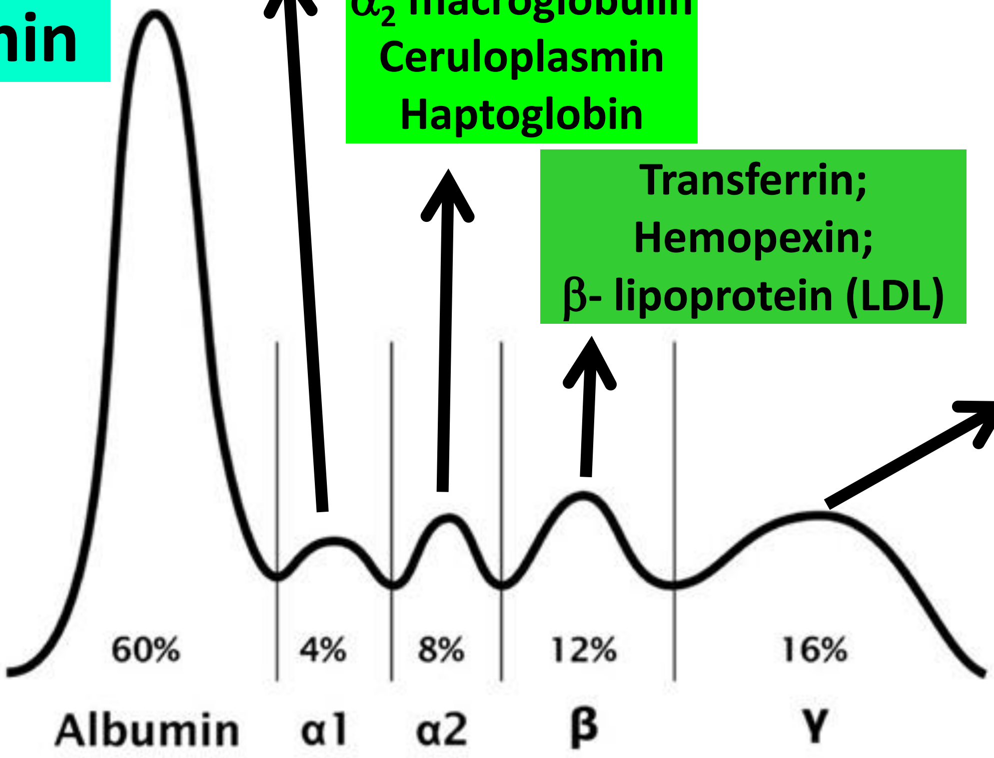
α_1-antitrypsin
α-fetoprotein
Transcortin; RBP

α_2 macroglobulin
Ceruloplasmin
Haptoglobin

Transferrin;
Hemopexin;
β- lipoprotein (LDL)

Immunoglobulins:
IgG, IgM, IgA, IgD, IgE

Synthesized by plasma cells (activated B-lymphocytes)



Synthesized by liver

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0318; 0321; 0322

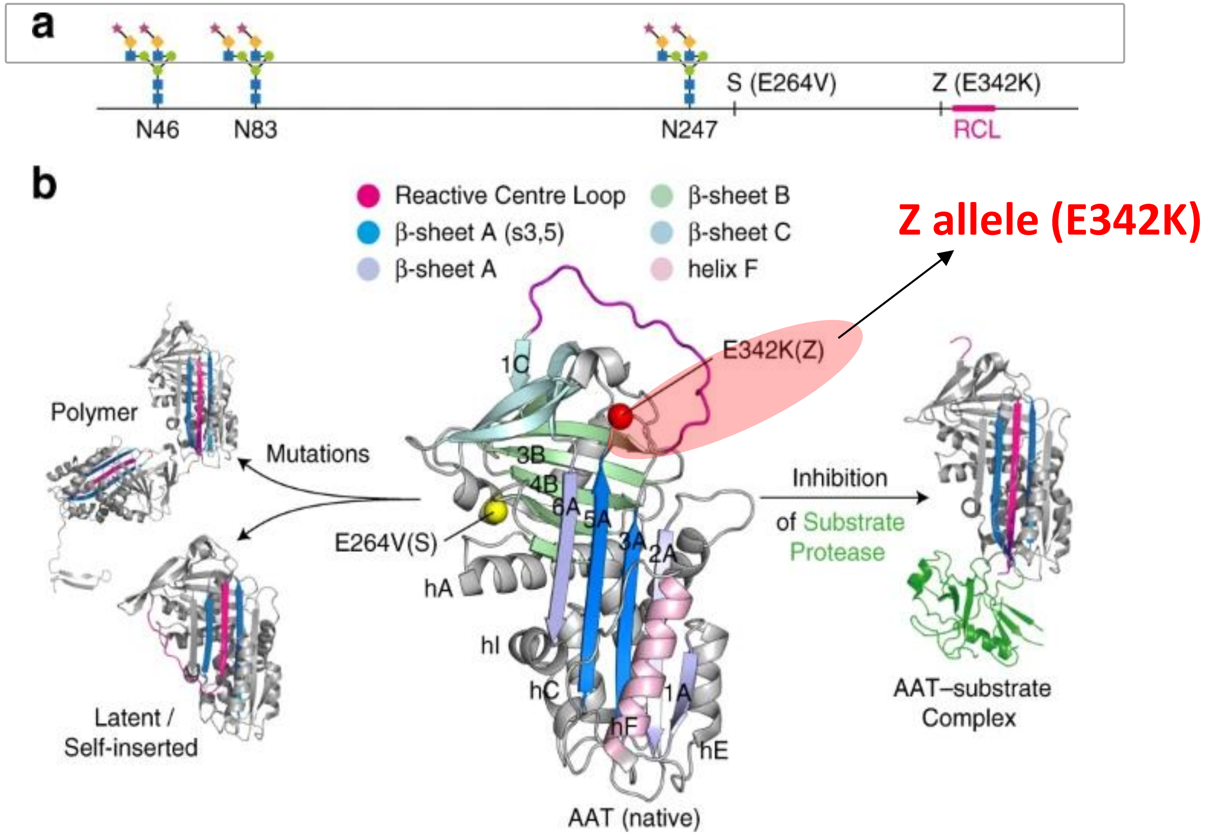
α_1 antitrypsin (Alpha 1-antitrypsin; AAT)

- ***SERPINA1*** gene encodes AAT
- Inhibitor of neutrophil elastase
  - Neutrophils release elastase as part of immune response
  - AAT protects **lung tissue** from neutrophil elastase
- AAT synthesized in LIVER and secreted into circulation
- AAT exists in polymorphic forms (Allelic heterogeneity)
  - **M** allele: Functional (Normal) allele
  - **Z allele:** Most frequent pathogenic allele **(Most severe)**
  - **S allele:** Less frequent deficiency allele
- Homozygous for Z allele (**PiZZ**): High risk of **pulmonary and liver disease**
- Heterozygous for Z allele (PiM**Z;** Pi**SZ**): Higher risk of pulmonary disease in smokers
- **Smoking** causes oxidation of AAT (and reduces its activity)

Alpha 1-antitrypsin: Polymorphic Alleles

Alleles of *SERPINA1* gene: **M**, **S**, **Z** (about 100 variants known)

- **M: Functional (Normal) allele**
- **Z : Most frequent pathogenic allele (Most severe)**
- **S: Less frequent deficiency allele**



Possible Genotypes:

(Pi: Protease inhibitor)

PiMM

PiM**S**

Pi**SS**

PiM**Z**

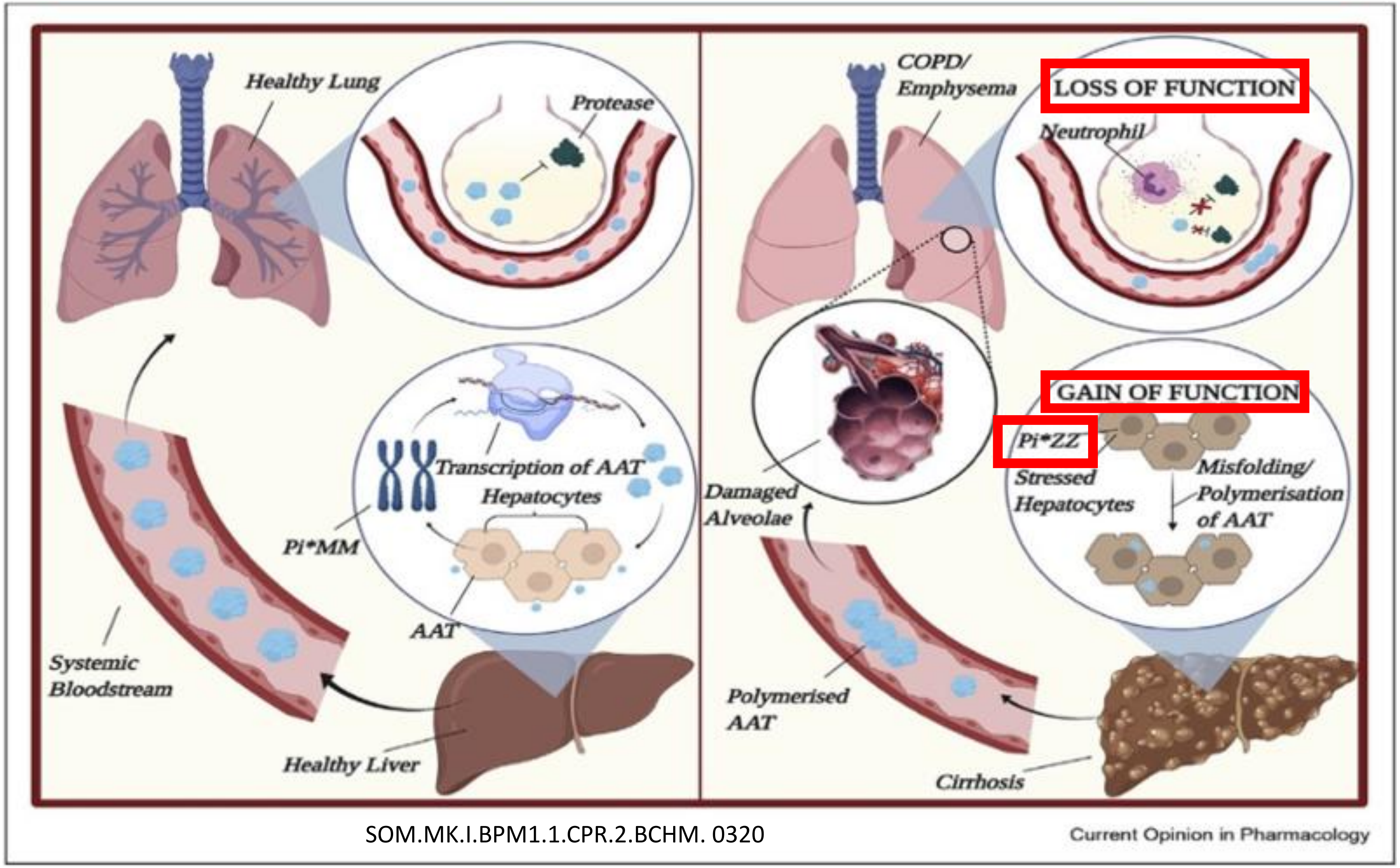
Pi**SZ**

PiZZ (Most severe)

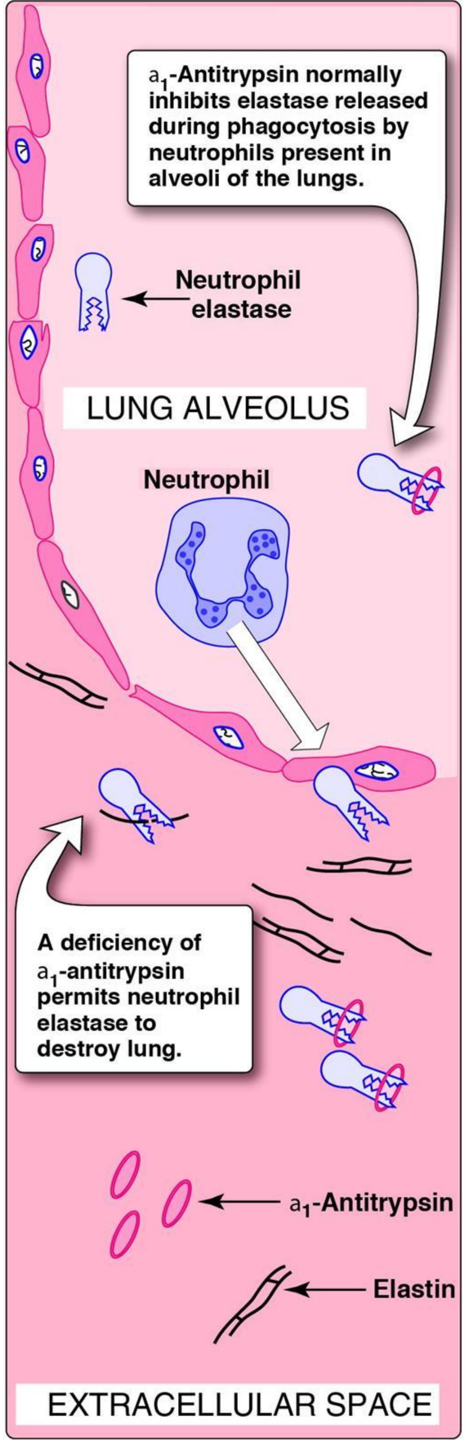
Jagger et et al. 2020, Nature Communications. High-resolution ex vivo NMR spectroscopy of human Z α1-antitrypsin

Lung pathology: Loss of function
Liver pathology (PiZZ): Gain of function

Normal AAT **Alpha 1-antitrypsin deficiency**

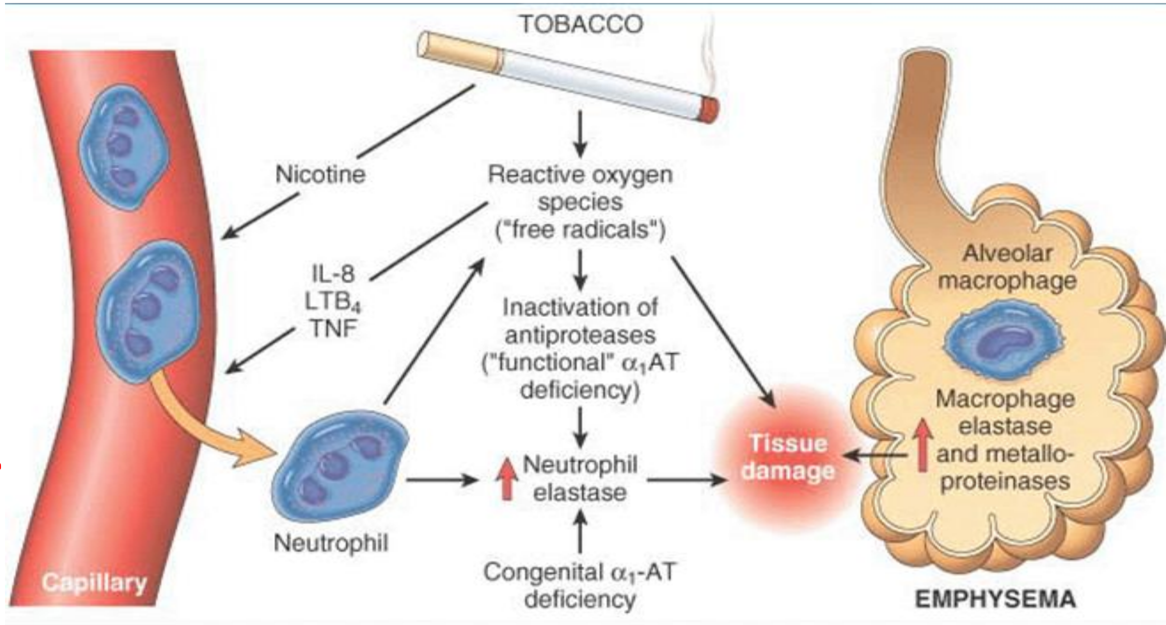


SOM.MK.I.BPM1.1.CPR.2.BCHM. 0320

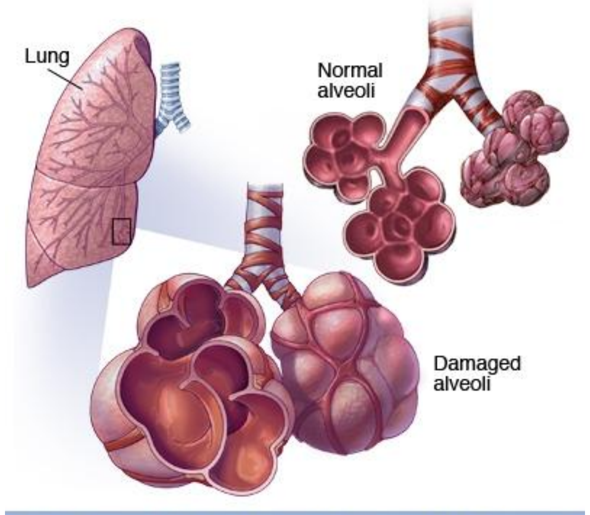


Lung pathology: Loss of function

- **Lung pathology: (PiZZ, PiSZ, PiMZ, PiSS)**
  - **"Loss of function" of AAT → Unopposed neutrophil elastase activity → Destruction of elastic tissue of alveoli (lung) → COPD and Emphysema**
  - **Made worse by smoking**
  - Age of onset: **30-50 years**
  - Respiratory symptoms: Dyspnea, cough….
- Management: Intravenous augmentation therapy with pooled human alpha1-antitrypsin slows progression of lung disease

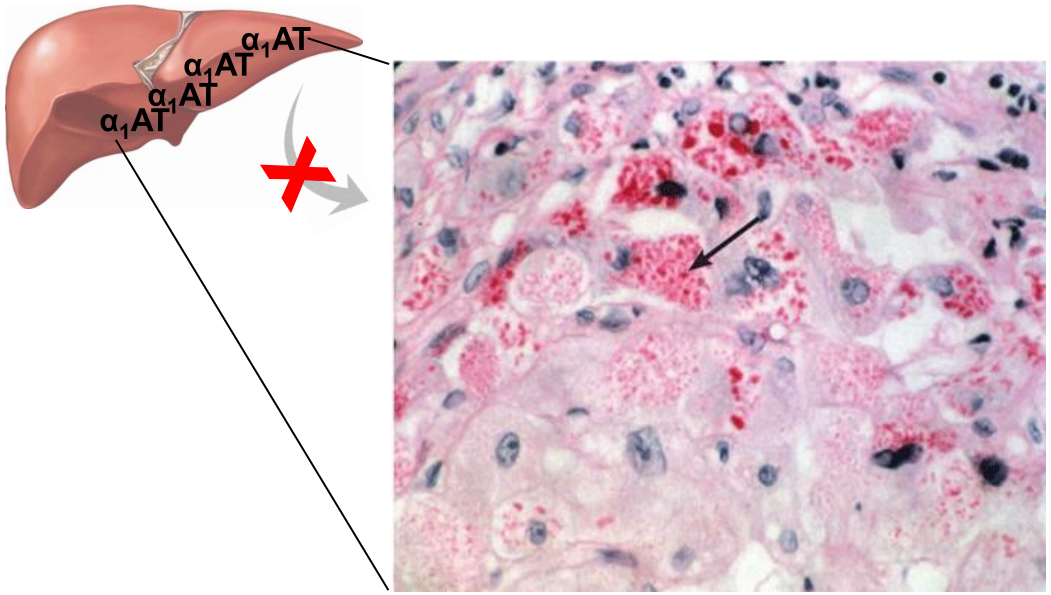


Pathogenesis of emphysema

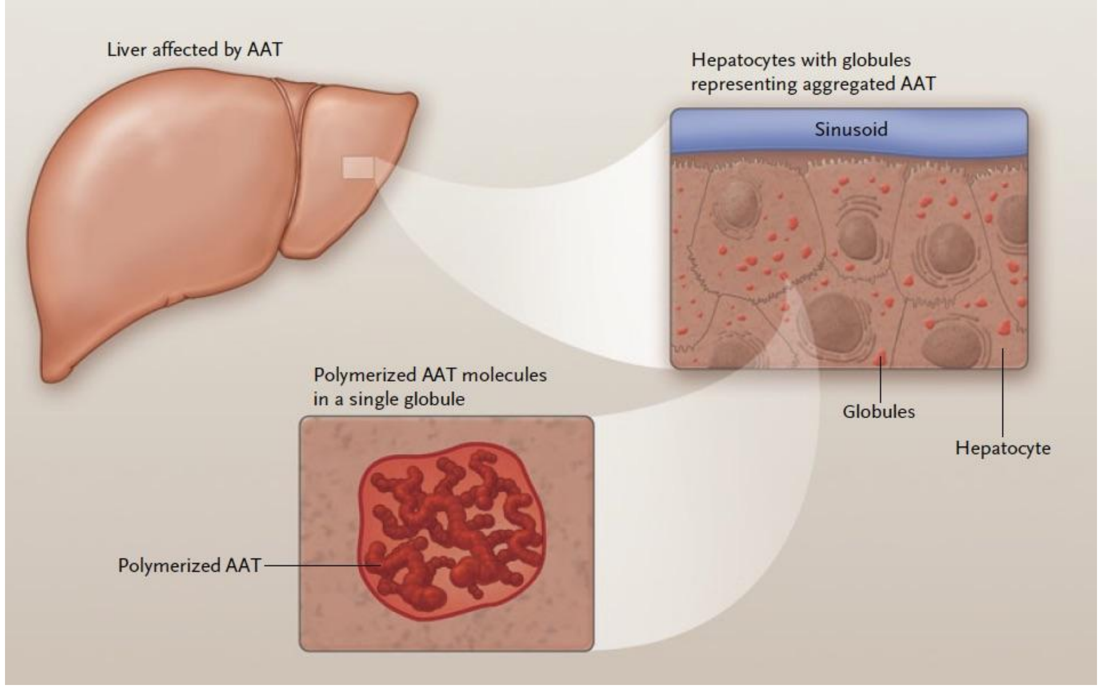


α1AT Deficiency: Effects on Liver (Gain of function)

- 'Misfolded' protein (**PiZZ**) accumulates in **Liver cells** → Hepatocyte damage → Cirrhosis **(attainment of novel function; "Gain of function")**
  - Seen only in some patients with **PiZZ** genotype



Periodic acid-Schiff staining

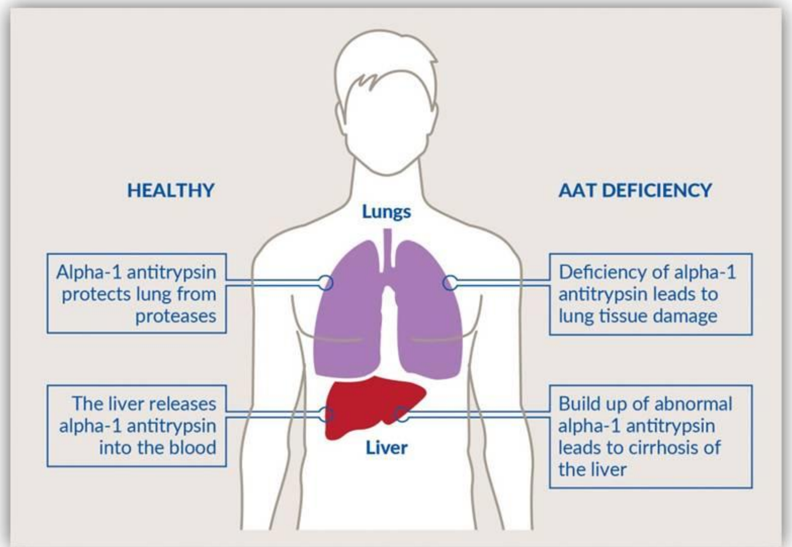


Liver cells show eosinophilic inclusion bodies (arrow) contain polymerized, unsecreted α1AT

Stoller, 2016, Cleveland Clinic Journal of Medicine. Alpha-1 antitrypsin deficiency: An underrecognized, treatable cause of COPD

AAT Deficiency illustrates several genetic principles

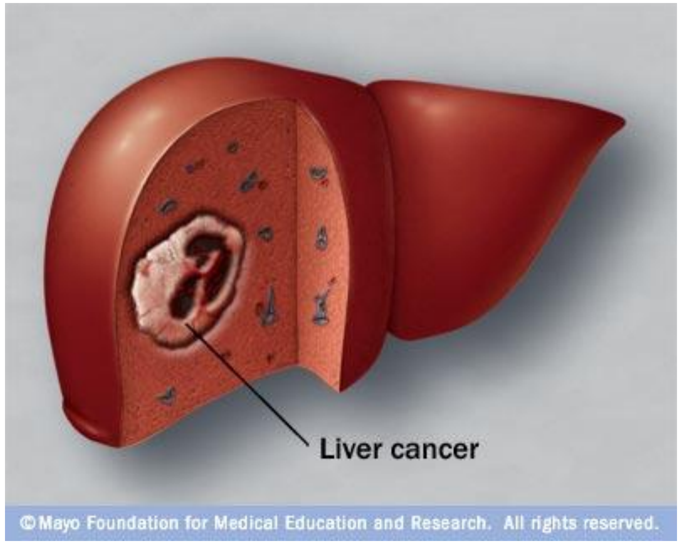
- Autosomal recessive; Codominant expression
- Allelic heterogeneity (**M**, **Z** and **S** alleles)
- Lung pathology due to “**loss of function**”
- Liver pathology due to attainment of a “**novel function**” (“**Gain of function**”)
- Single gene disorder heavily influenced by environment and behavior (air pollution and **smoking**)
- Augmentation therapy (Infusion of AAT protein)
- Screen first degree relatives of index patient



https://medlineplus.gov/genetics/condition/alpha-1-antitrypsin-deficiency/
Alpha-1 Antitrypsin Deficiency - causes, symptoms, diagnosis, treatment, pathology – YouTube
A Closer Look at Alpha-1 Antitrypsin Deficiency: Peggy’s Story - YouTube

Alpha-fetoprotein (AFP)

- Alpha 1-globulin
- Abundant in fetus (may function as fetal albumin)
- Elevated AFP levels: **Tumor marker** for **liver cancer**, ovarian or testicular cancer
- Maternal serum screening (Pregnancy) may identify fetal anomalies
  - **Elevated maternal serum AFP**: Neural tube defects (Spina bifida and anencephaly)
  - **Low maternal serum AFP**: Trisomy 21 (Down syndrome)
  - Not confirmatory but require further investigations

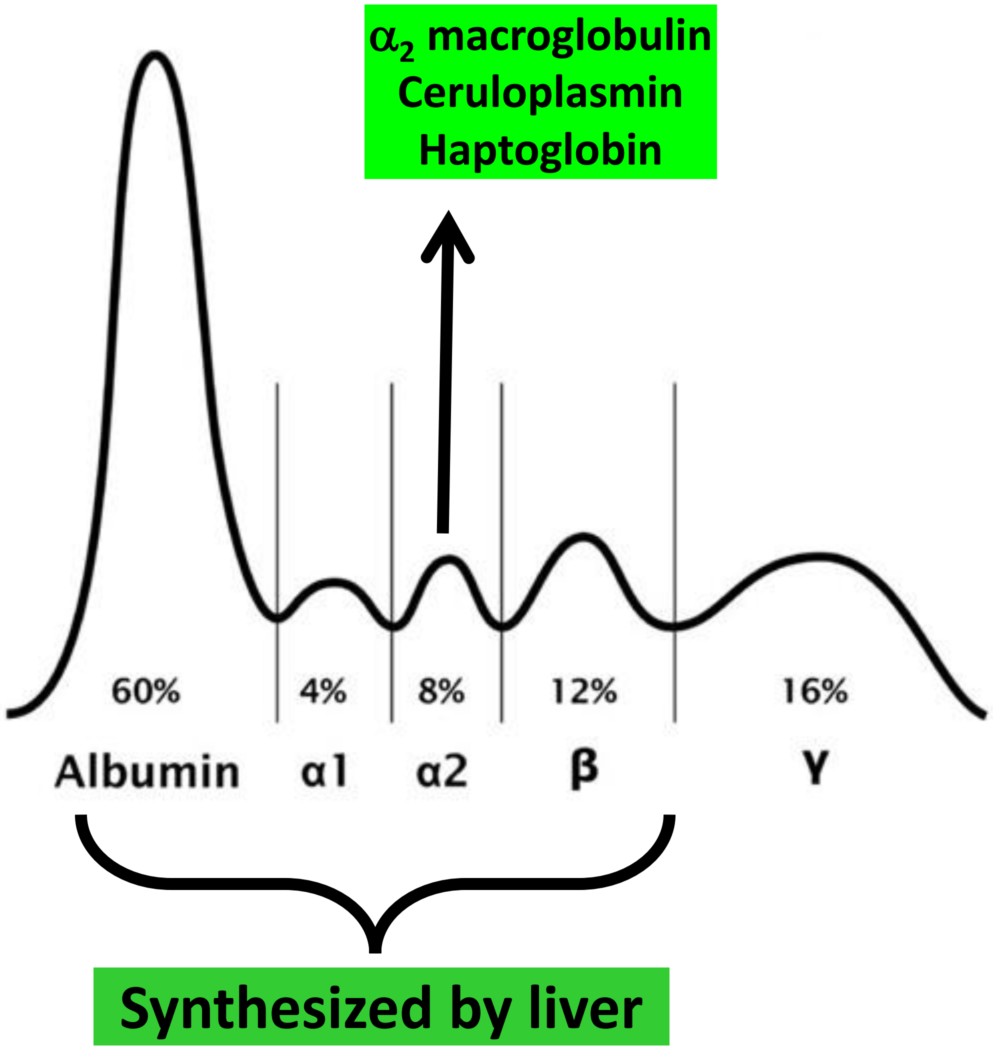


SOM.MK.I.BPM1.1.CPR.2.BCHM. 0318

Transcortin and retinol-binding protein (RBP)

- Both are **negative acute phase proteins** (Reduced synthesis following inflammation)
- Transcortin transports cortisol (steroid hormone)
- Retinol-binding protein transports retinol (Vitamin A)

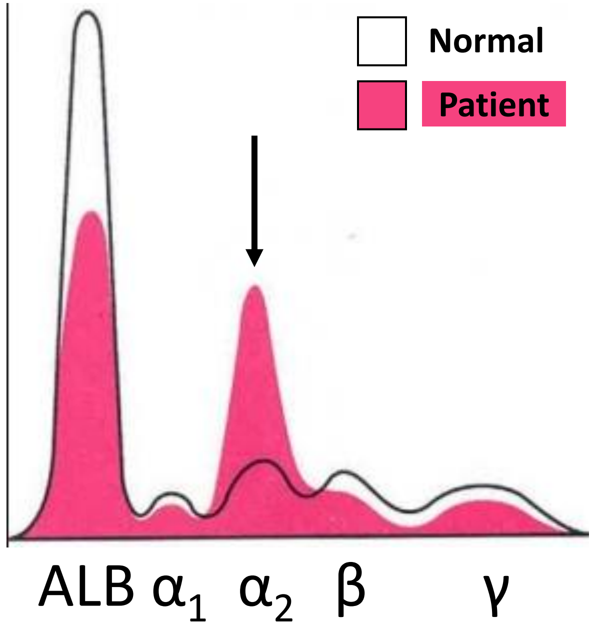
Proteins in the α-2 globulin fraction



Alpha2-macroglobulin (α2-macroglobulin)

- Largest of the serum proteins
- Inhibits many proteases like plasmin and thrombin
- Levels elevated in **nephrotic syndrome**
  - Loss of glomerular basement membrane → loss of albumin → Stimulates synthesis of ALL proteins by liver
  - α2-macroglobulin levels elevated (Large protein retained)
  - Serum cholesterol elevated (High LDL-cholesterol)

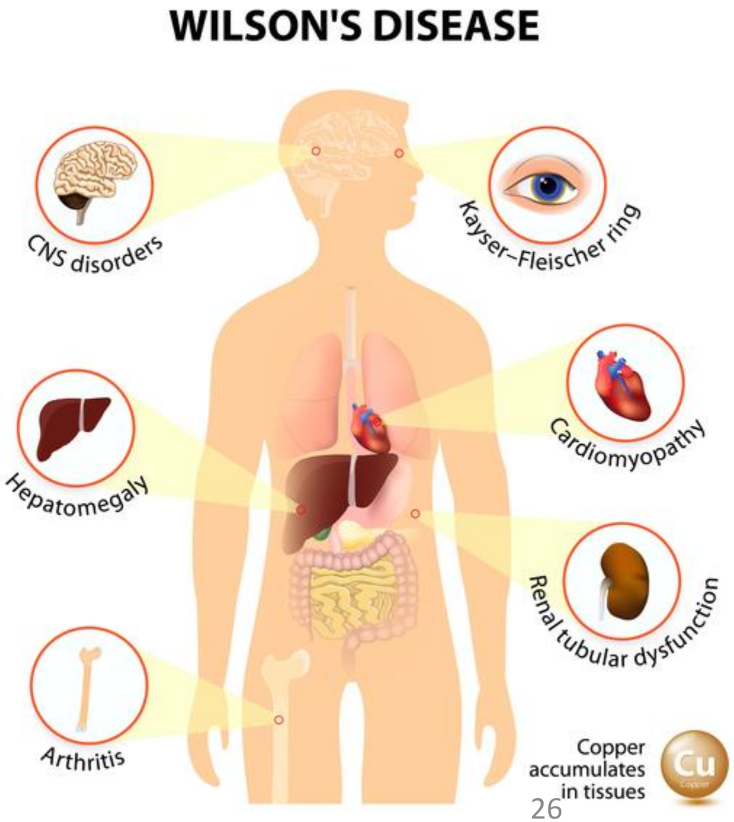
Nephrotic syndrome: Serum protein electrophoresis



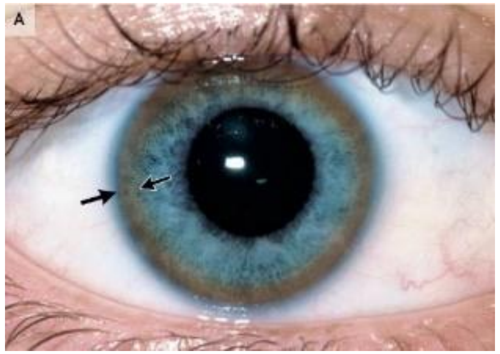
Ceruloplasmin

- Copper containing plasma protein
- Synthesized and secreted by liver
- Helps iron mobilization: Ceruloplasmin has ferroxidase activity; Oxidation of Fe^{2+} (ferrous) to Fe^{3+} (Ferric); Ferric ions taken up by transferrin
- **Low ceruloplasmin levels: Wilson's disease** (copper overload disorder): **Liver; Corneal ring; basal ganglia involvement** (Involuntary movements; mood changes)
  - Inability to incorporate copper into ceruloplasmin
  - Low excretion of copper in bile

Fe^{2+} (Ferrous) $\xrightarrow{\text{Ceruloplasmin (Ferroxidase)}}$ Fe^{3+} (Ferric) → Incorporated into transferrin

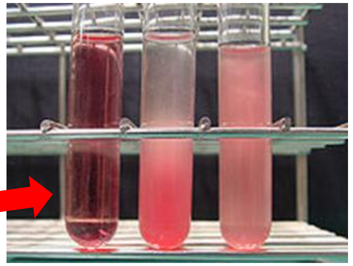


Kayser-Fleischer ring

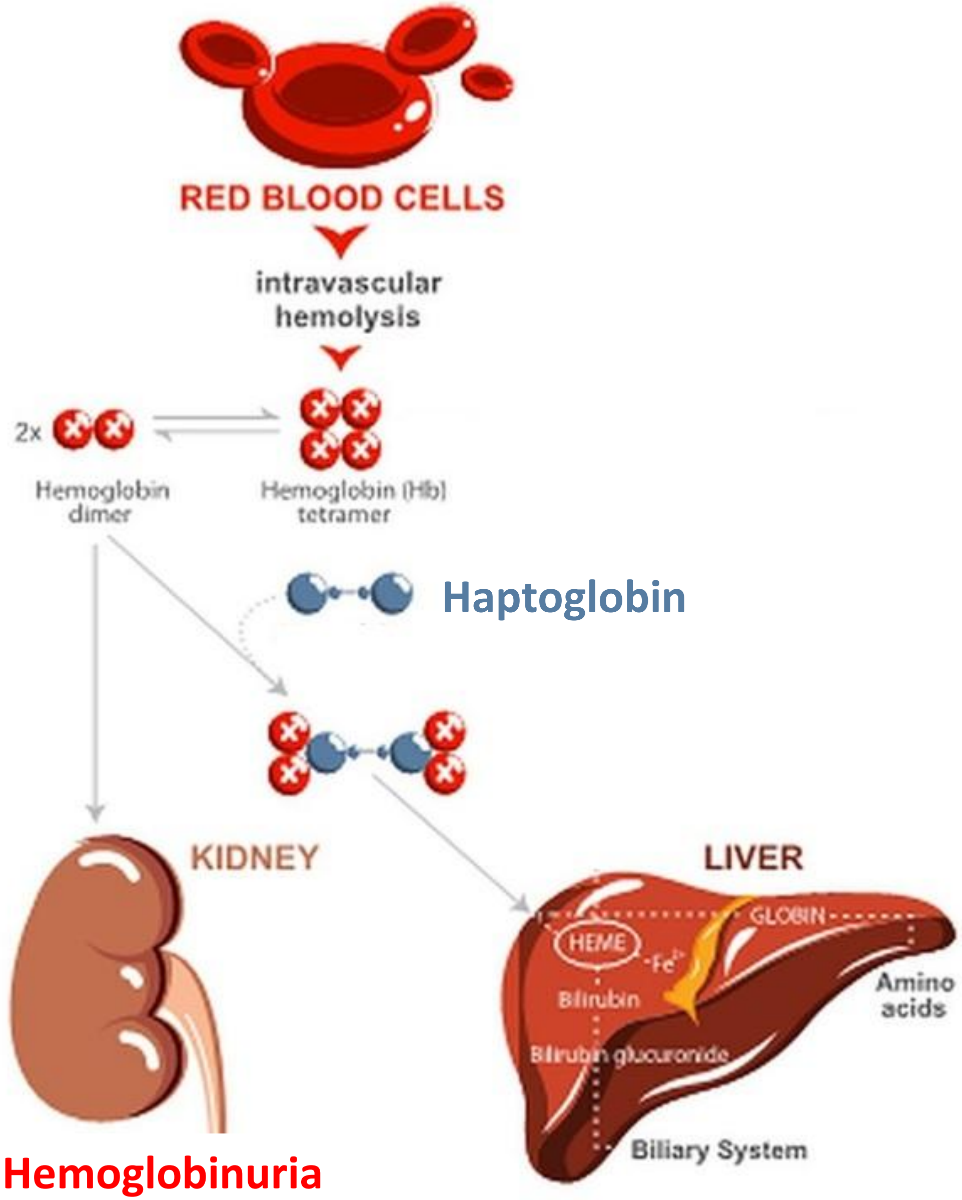


Haptoglobin

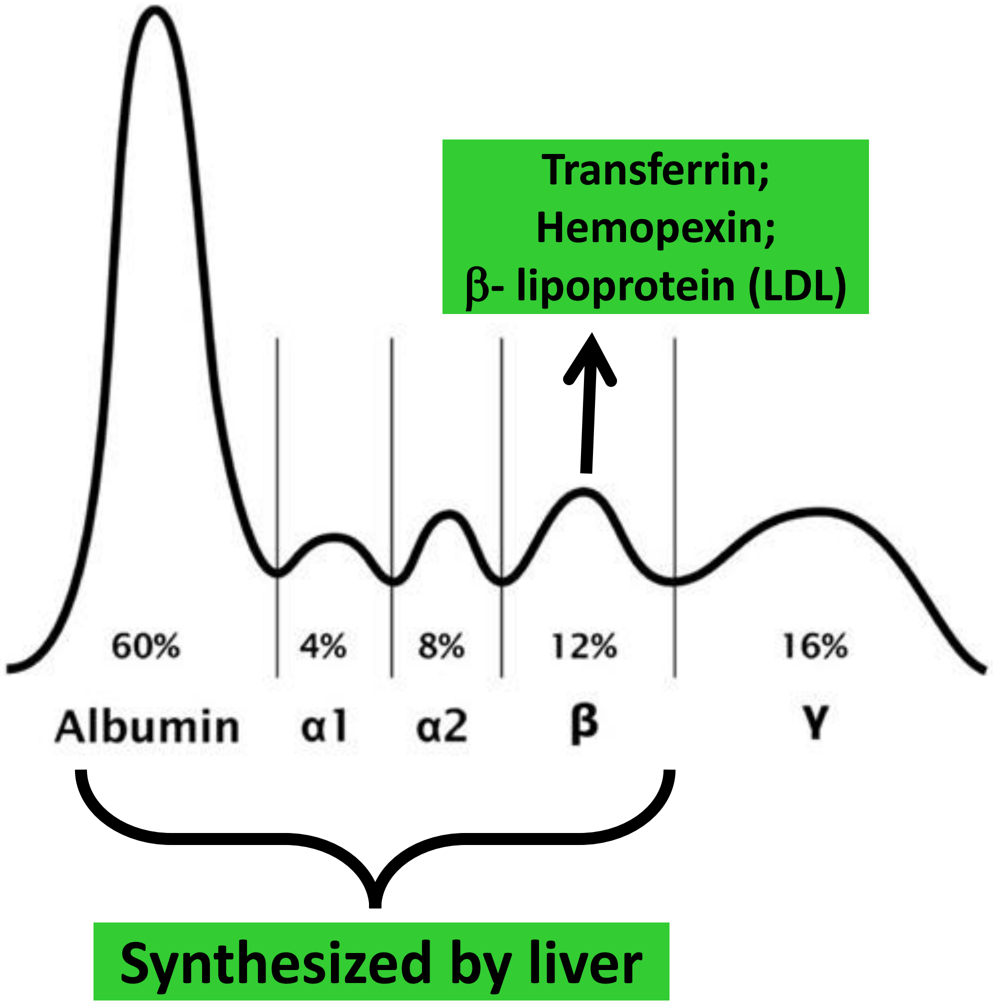
- Binds **hemoglobin** in circulation (hemolysis)
- Haptoglobin-hemoglobin complex → Taken up by LIVER
- Haptoglobin reduces loss of hemoglobin in urine
- Conserves hemoglobin: Amino acids and iron
- **Low Haptoglobin levels following intravascular hemolysis**
  - **Haptoglobin levels: Indicator hemolysis**



Hemoglobinuria

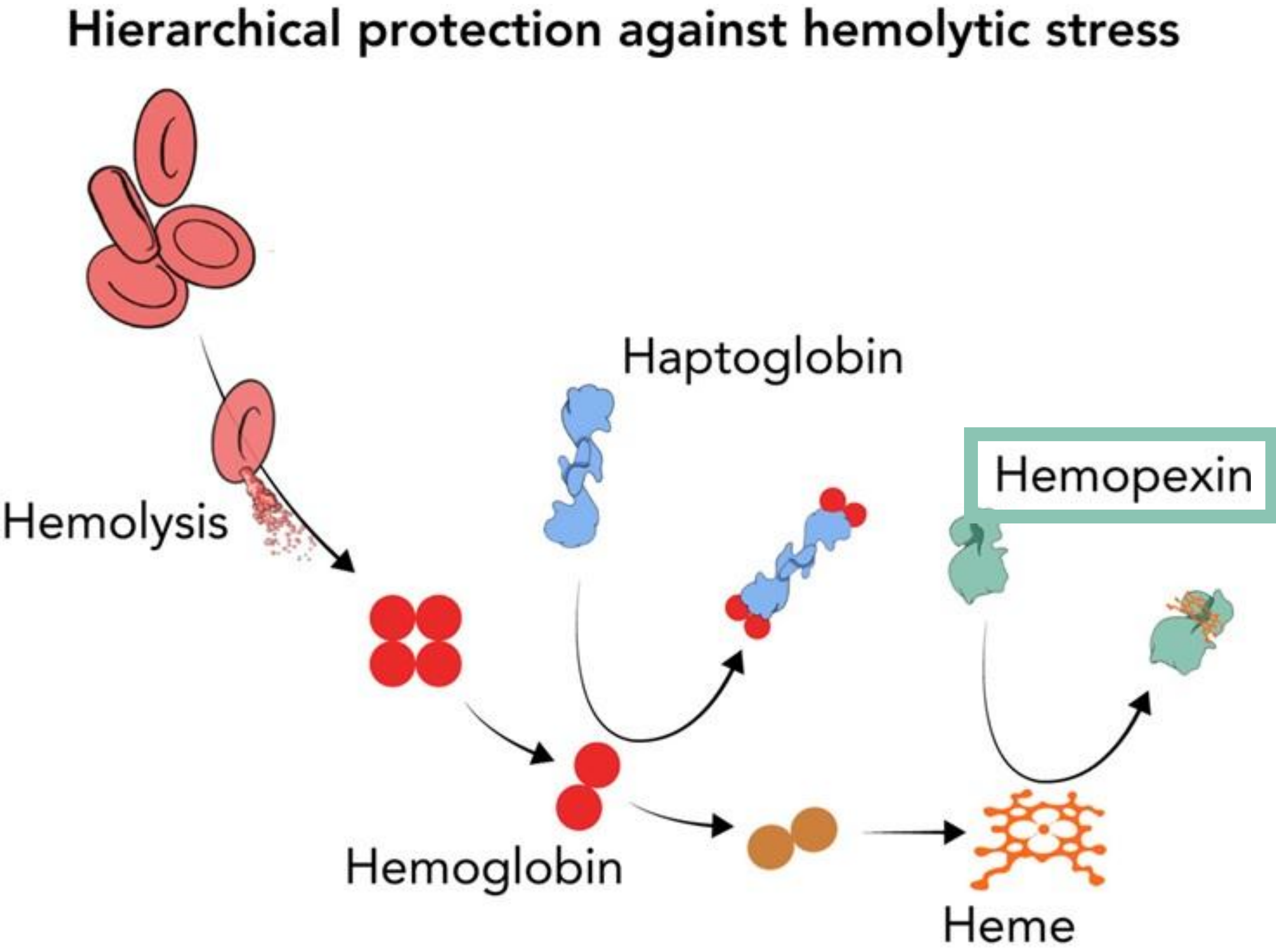


Proteins in the β-globulin fraction



Hemopexin

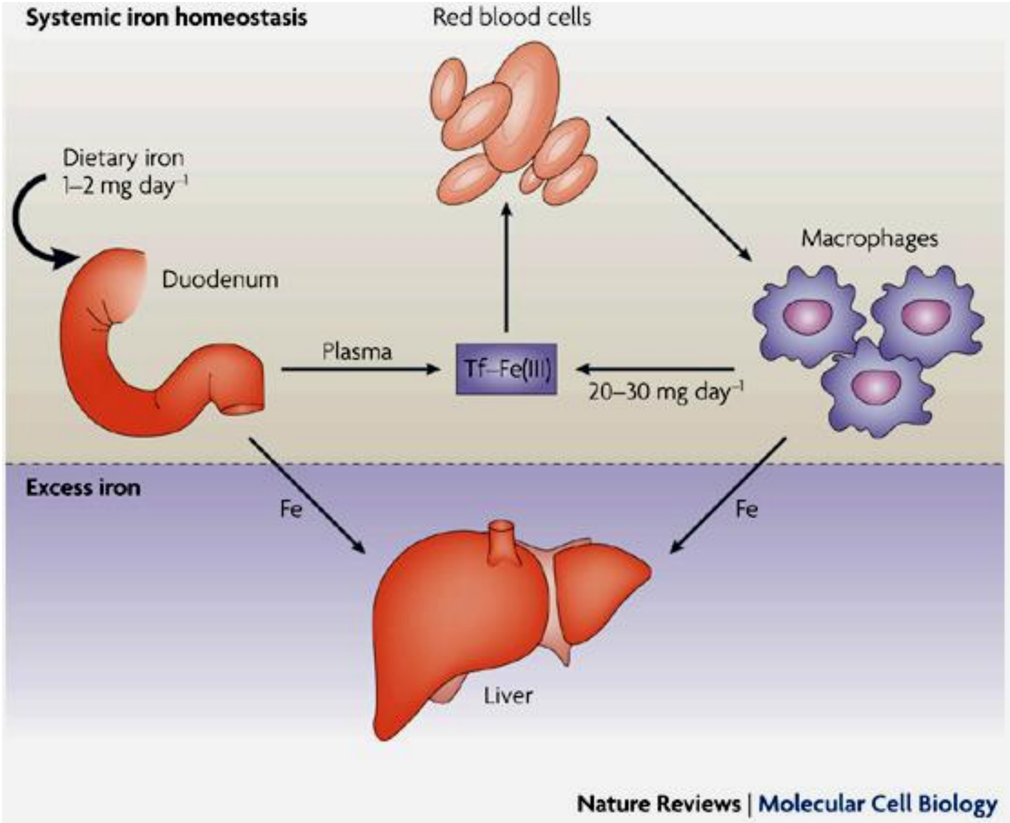
- Binds to **free heme** following hemolysis
- Hemopexin-heme complex is taken up by liver
- Conserve iron
- Protects from heme-induced cell damage
- **Haptoglobin binds to hemoglobin**



SOM.MK.I.BPM1.1.CPR.2.BCHM. 0322

Transferrin

- Transport protein for iron (Ferric)
- Transports iron between intestine, liver, bone marrow and reticuloendothelial system (spleen and macrophages)
- Each transferrin can bind 2 Fe^{3+} atoms
- Normally 1/3rd of binding sites on transferrin are filled (saturated) with iron
- Low serum iron and low transferrin saturation: iron deficiency anemia
- High serum iron and high transferrin saturation: Iron overload



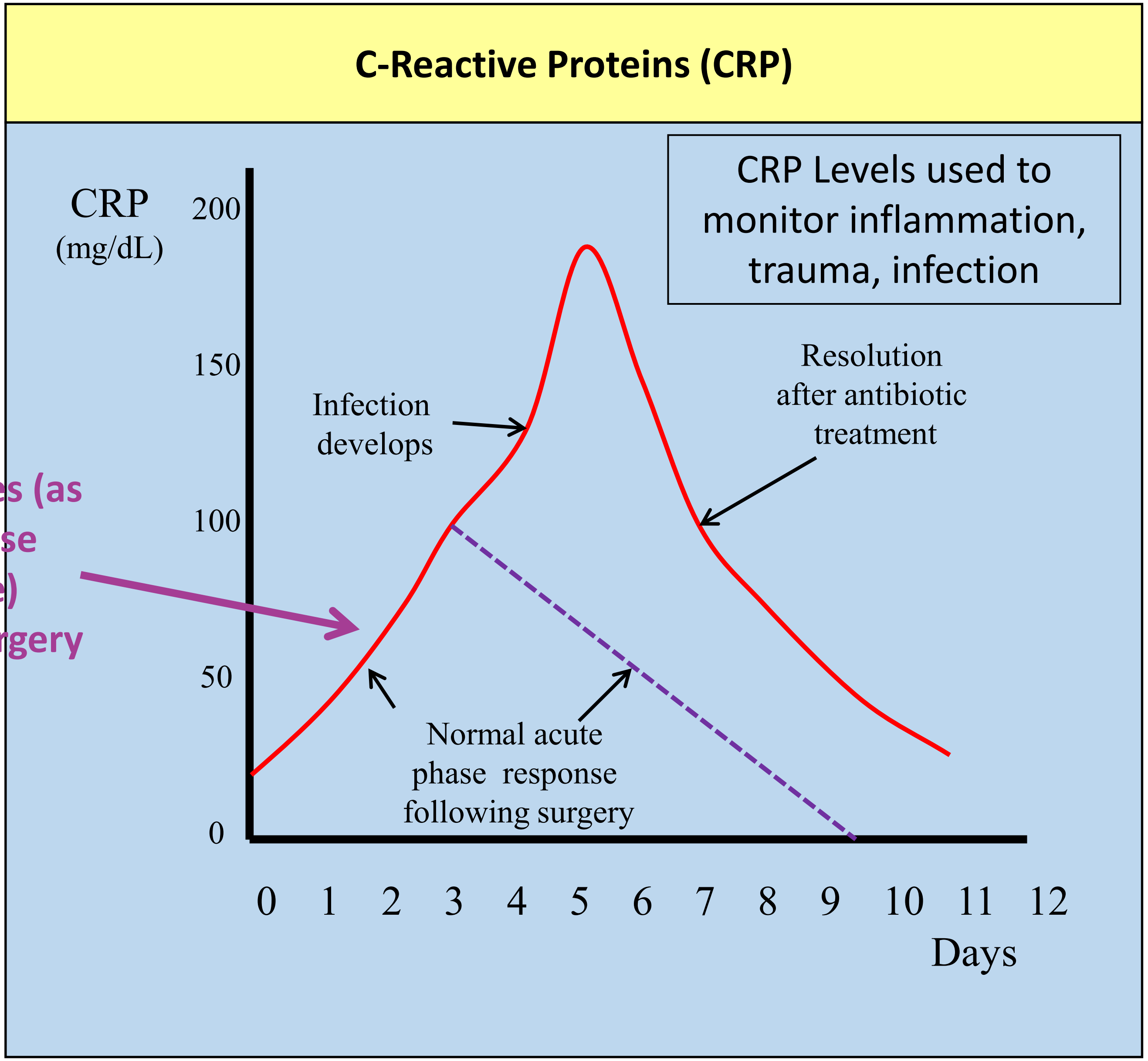
Low density lipoprotein (LDL; beta-lipoprotein)

- Migrates to beta-globulin region
- Elevated LDL in
  - Familial hypercholesterolemia (LDL-receptor mutation; Type IIA hyperlipidemia)
  - Nephrotic syndrome (Elevated serum total cholesterol and LDL-cholesterol levels)
- Increases risk of atherosclerotic cardiovascular disease

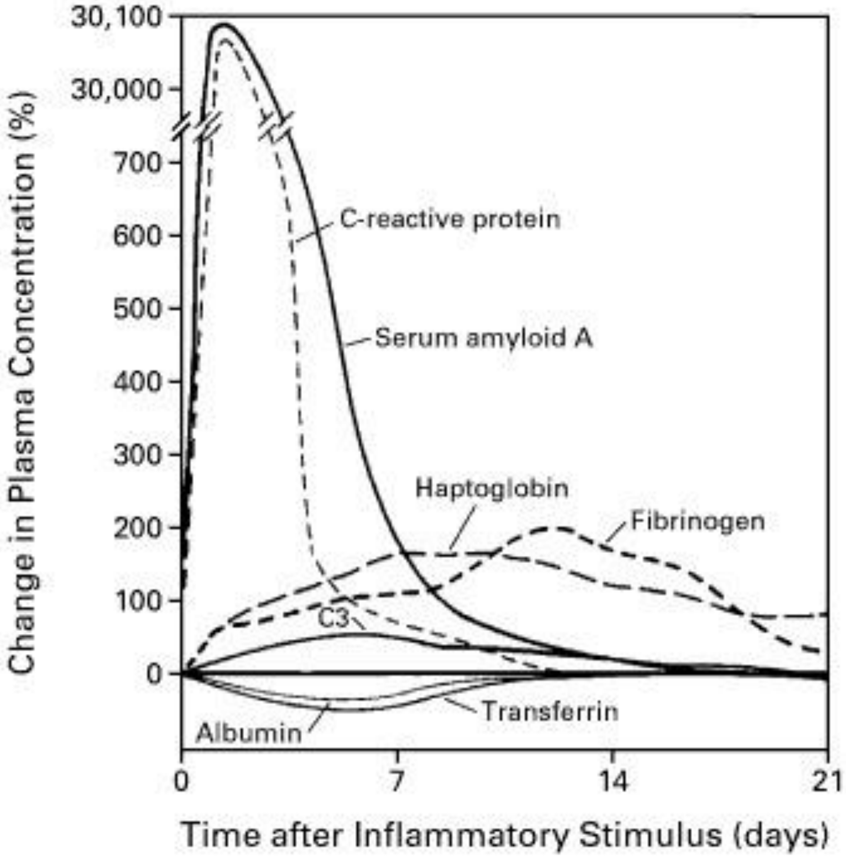
Acute phase proteins (Acute phase reactants)

- Group of proteins increased in any inflammatory disorder (**non-specific**)
- Increased **hepatic** synthesis of these proteins in inflammation (Infections, burns, crush injuries, surgery….)
- **Cytokines** (inflammatory mediators) stimulate **hepatic synthesis**
- **Positive Acute phase reactants**:
  - Levels increase in inflammation
  - **C-reactive protein (CRP),** ceruloplasmin, haptoglobin, alpha1-antitrypsin, fibrinogen
  - CRP levels measured to monitor **progress** of inflammatory reaction
- Negative acute phase reactants:
  - Synthesis of some proteins decrease following an inflammatory response
  - Amino acids used for synthesis of proteins useful in inflammatory response
  - **Albumin**, transferrin

Acute phase proteins (Acute phase reactants): C-reactive protein



CRP increases (as acute phase response) following surgery

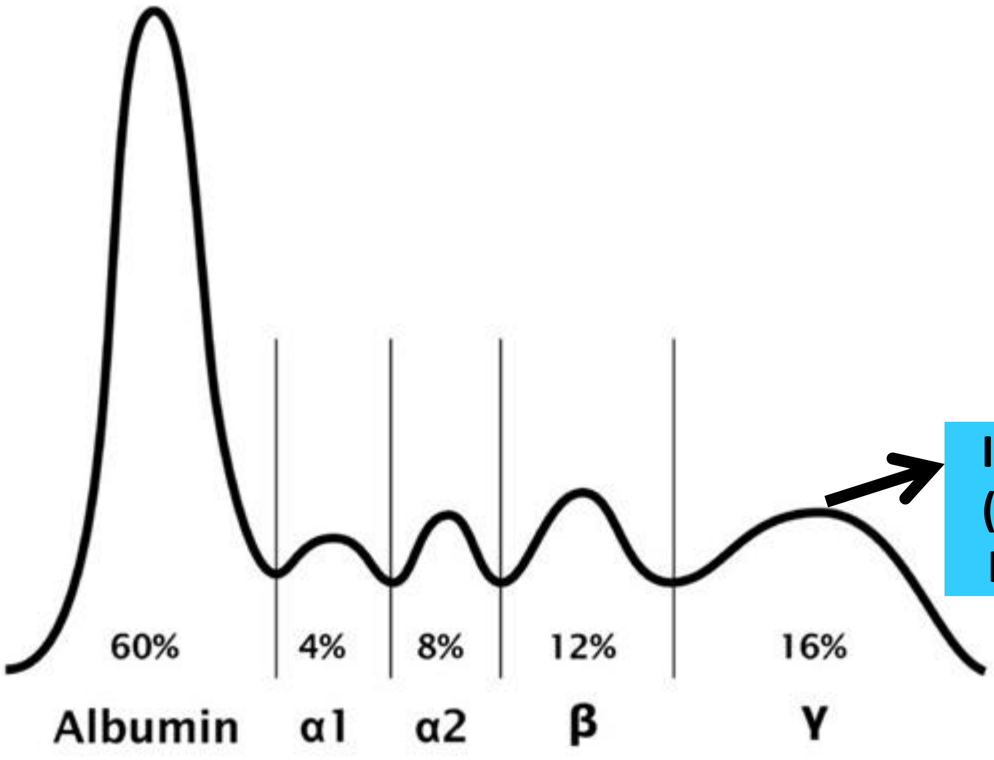


SOM.MK.I.BPM1.1.CPR.2.BCHM. 0324; 0325

Acute Phase Reactants (APR)

Positive acute phase reactants (Increased Synthesis)	Negative acute phase reactants (Decreased Synthesis)
C-Reactive Protein (CRP)	**Albumin**
α1 Antitrypsin *(α_1 globulin)*	Retinol binding protein *(α_1 globulin)*
Ceruloplasmin *(α_2 globulin)*	Transcortin *(α_1 globulin)*
Haptoglobin *(α_2 globulin)*	Transferrin *(β globulin)*
Hemopexin *(β globulin)*	
Immunoglobulins *(γ globulin)*	
Ferritin	
Complement proteins	

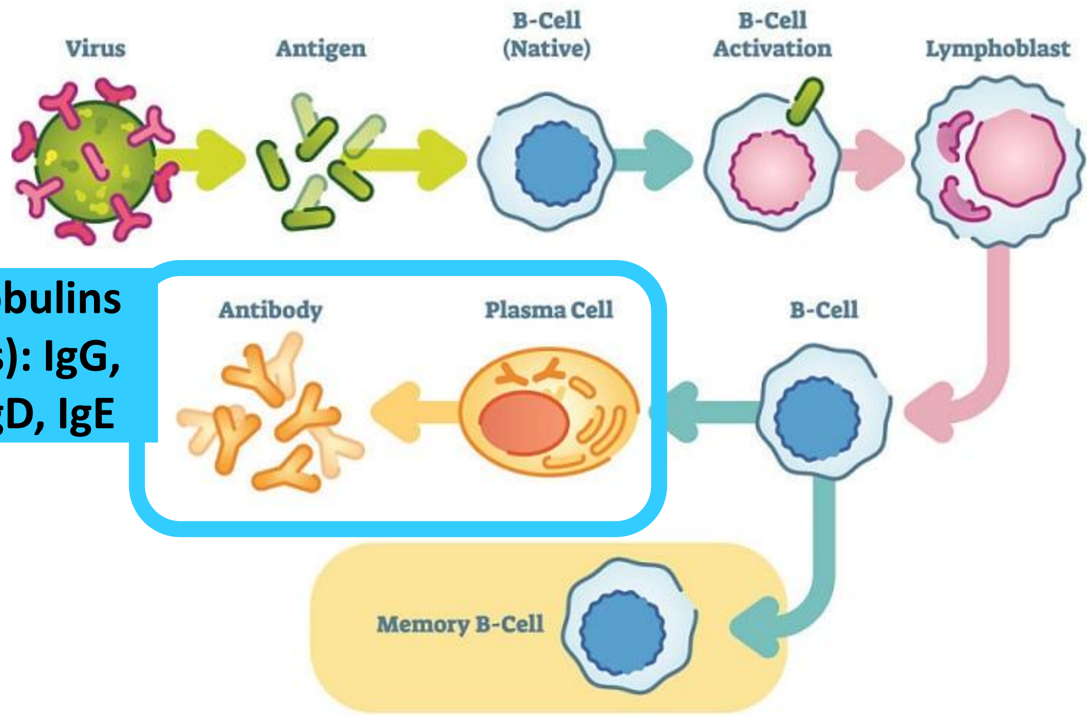
Proteins in γ globulin fraction (Immunoglobulins/Antibodies)



Synthesized by liver

Synthesized by plasma cells (Activated B-lymphocytes)

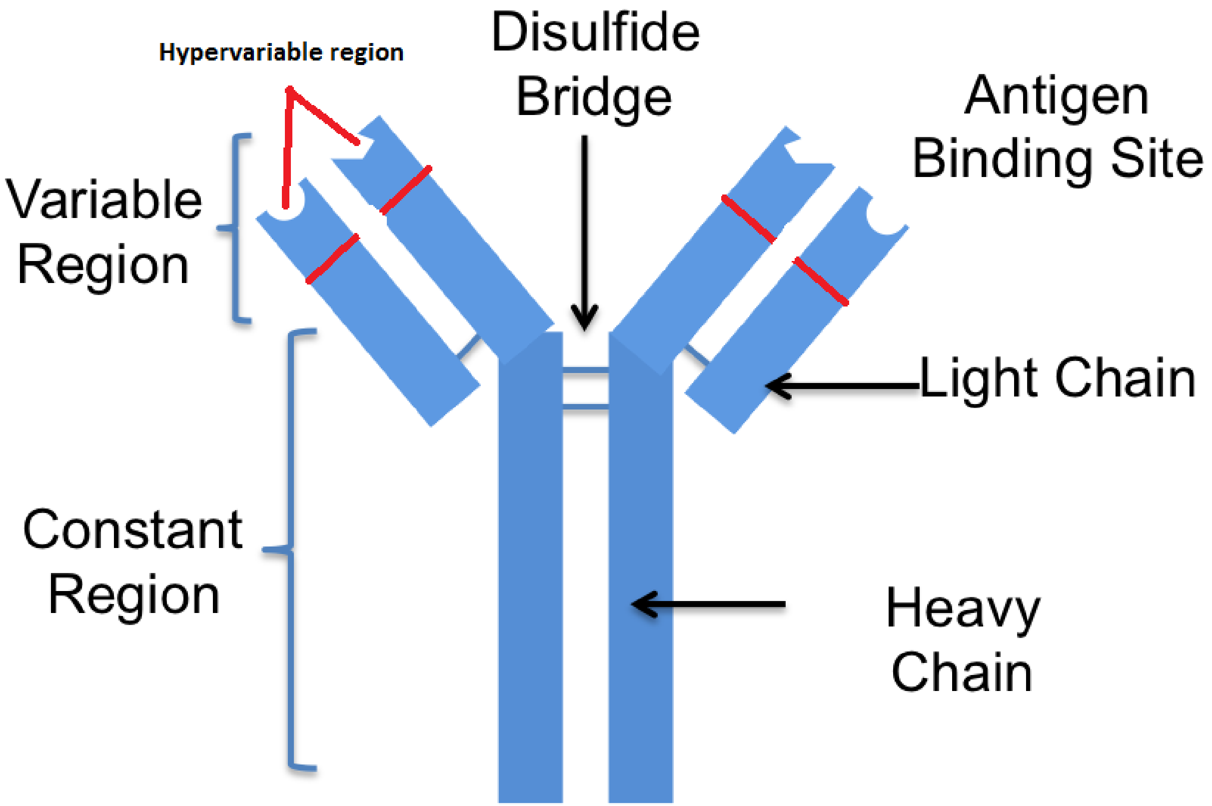
Immunoglobulins (Antibodies): IgG, IgM, IgA, IgD, IgE



Self-study

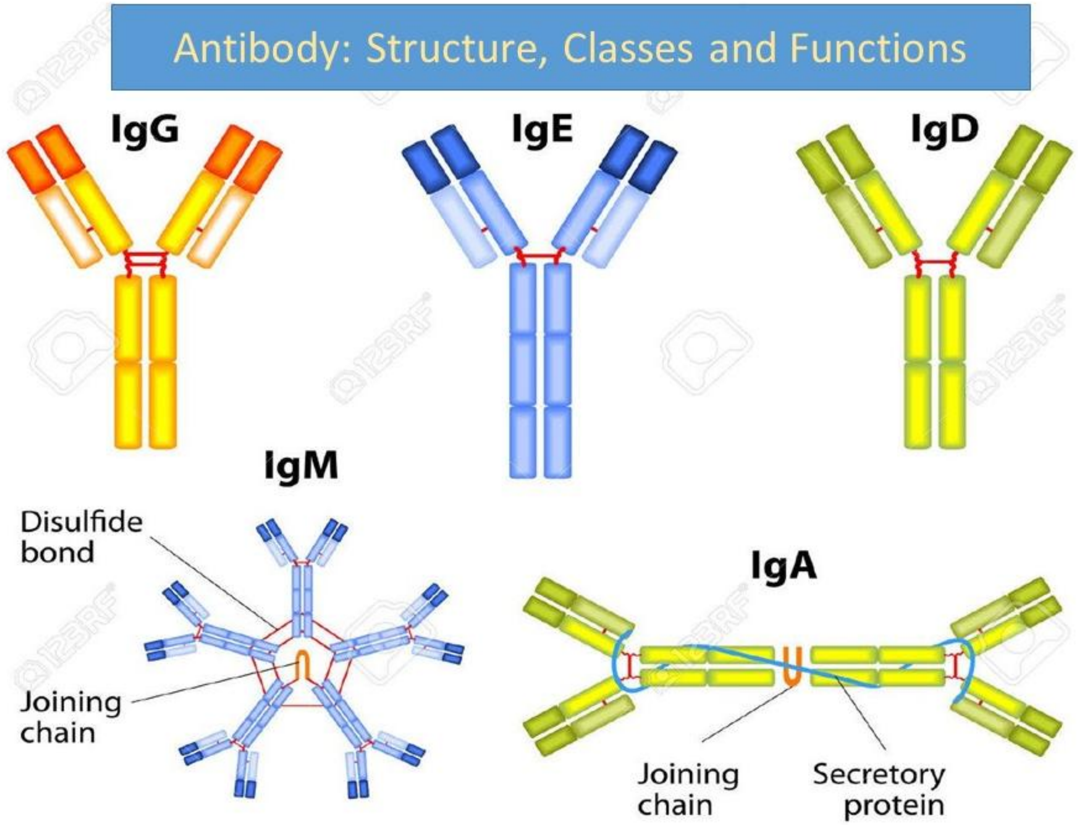
General structure of immunoglobulin

- Y-shaped molecule
- 2 Heavy chains and 2 light chains
- Contain two antigen binding sites
- Classified into 5 types based on Heavy chain



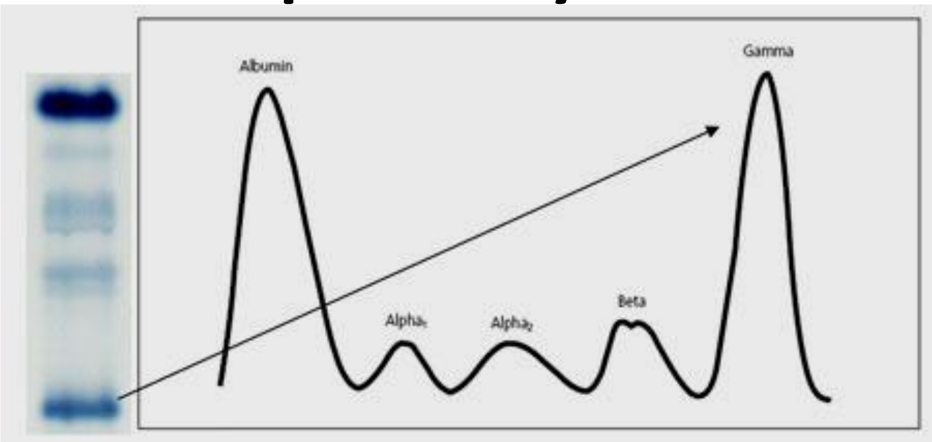
Immunoglobulin classes

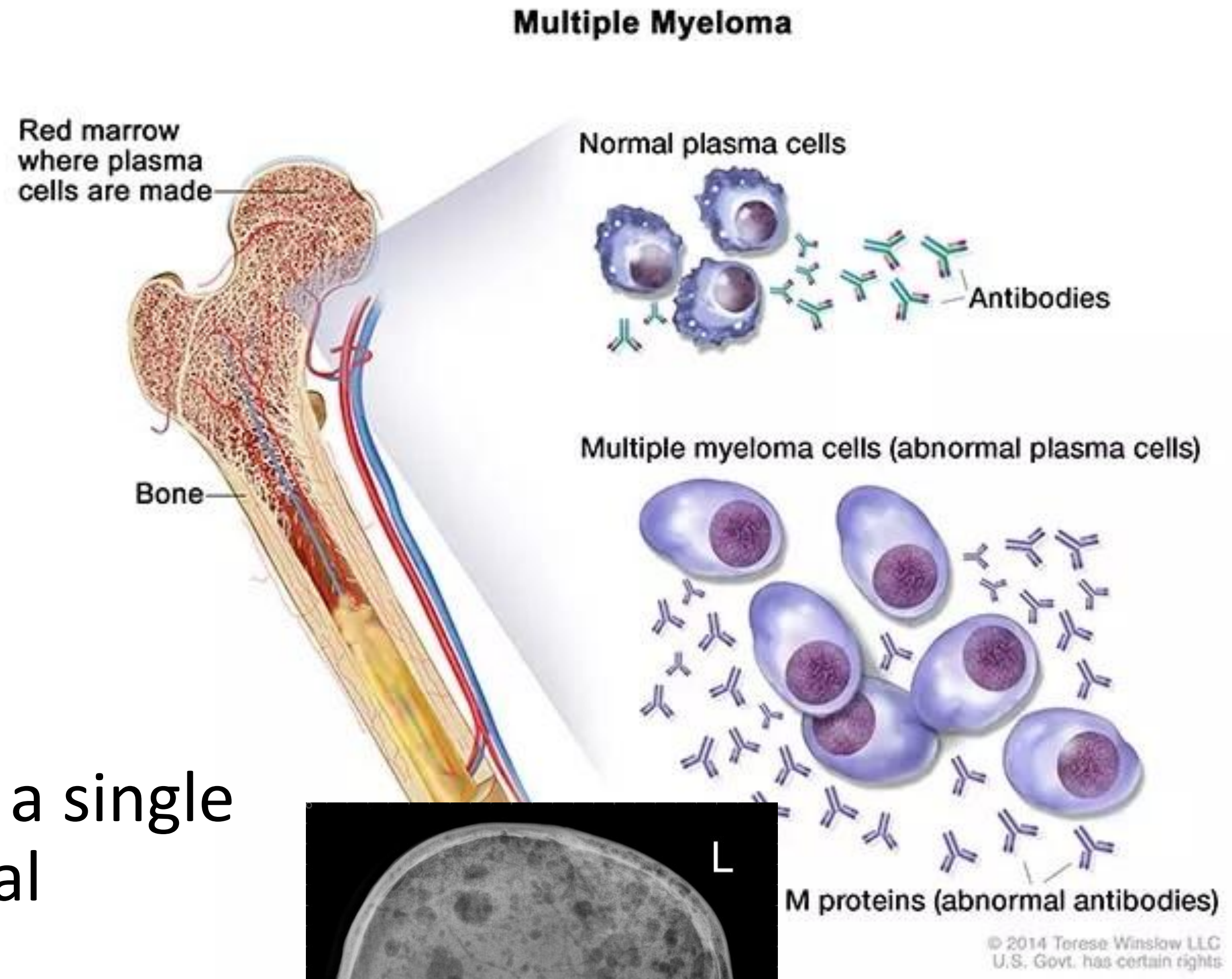
- IgM: First antibody produced in response to an antigen (Largest)
  - Pentamer
- IgG: Produced on repeated exposure to the same antigen (Most abundant)
  - IgG crosses placenta and confers passive immunity to fetus and newborn
- IgE: Secreted in an allergic response
- IgA: Secretory immunoglobulin; Dimer; Present in body secretions like saliva, breast milk…



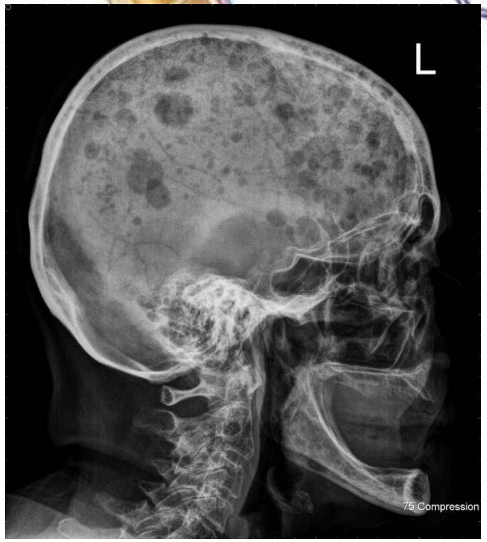
IgM: Pentamer

Multiple myeloma



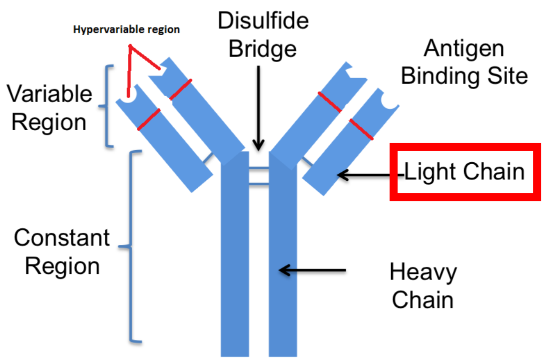


- **Plasma cell tumor →** production of a single type of immunoglobulin (monoclonal gammopathy)
- Affects bones and presents as fractures or bone pain (due to lytic lesions in bones)

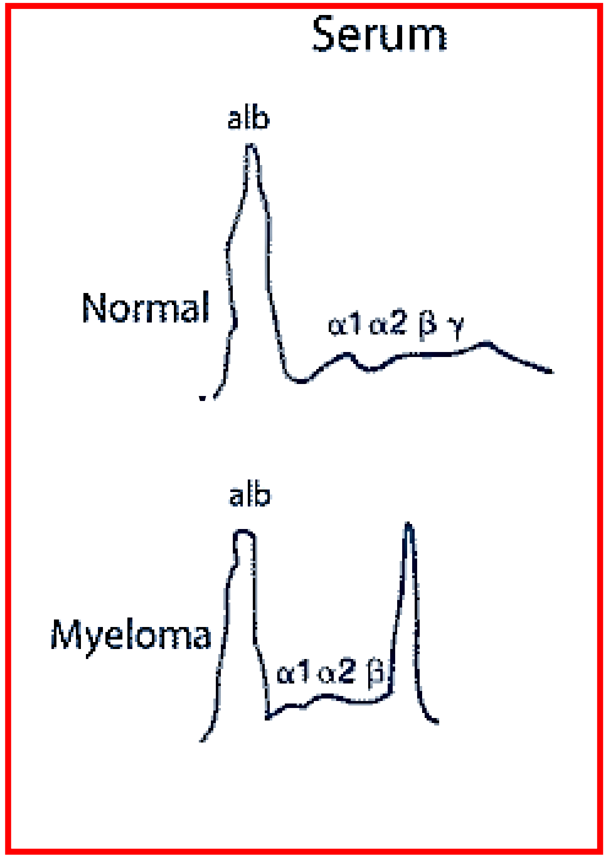


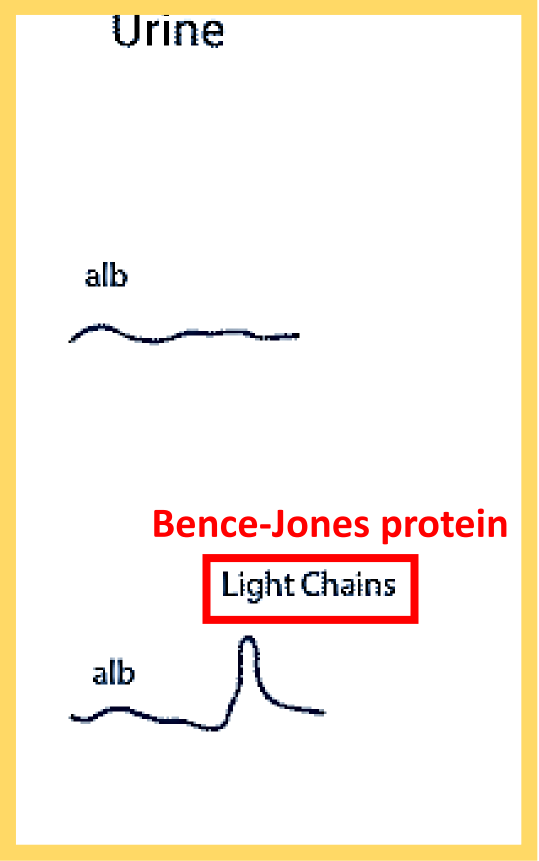
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0324; 0323

Multiple myeloma: Lab findings



- Serum protein electrophoresis: Monoclonal peak in gamma-globulin
- Urine findings: **Excess light-chains** of immunoglobulins found in urine **(Bence-Jones protein)**
  - Light chains are freely filtered
  - **Not detected by protein dipstick tests**
  - Urine electrophoresis





SOM.MK.I.BPM1.1.CPR.2.BCHM. 0323; 0326

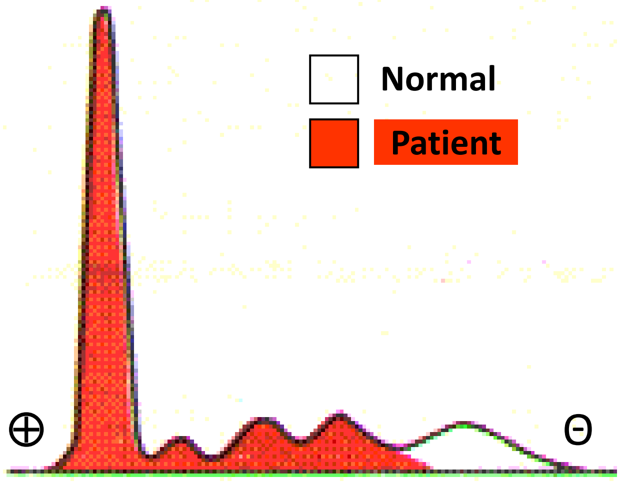
Case studies

- Answers in Practice questions!!

Case 1

A 6-month-old boy is brought to the emergency room because of difficulty breathing. His temperature is 104°C, respirations are labored, and pulse is rapid. He has a history of repeated skin and respiratory infections. He is admitted and treated with broad spectrum antibiotics and oxygen. Total serum protein levels were slightly below normal. Densitometry of serum protein electrophoresis is shown.

- Interpret the serum protein electrophoresis in this patient



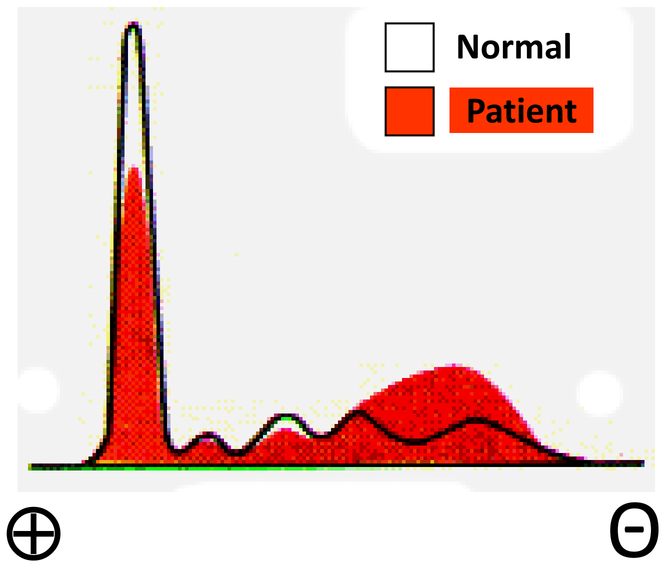
- Name the component proteins of this region:
- Findings are from a patient with:_________
- What are the normal functions of γ-globulins?

- Which is the immunoglobulin produced as the first line of defense to an antigen and which is produced on repeated exposure?

Case 2

A 60-year-old woman comes to the physician because of 3-week history of abdominal swelling and yellowish discoloration of eyes. She has a 15-year history of alcohol use disorder. Physical examination shows ascites and bilateral pitting edema. Densitometry of serum protein electrophoresis is shown.

- Identify the various bands in the densitometry pattern of serum protein electrophoresis
- Identify the bands that are abnormal in the patient

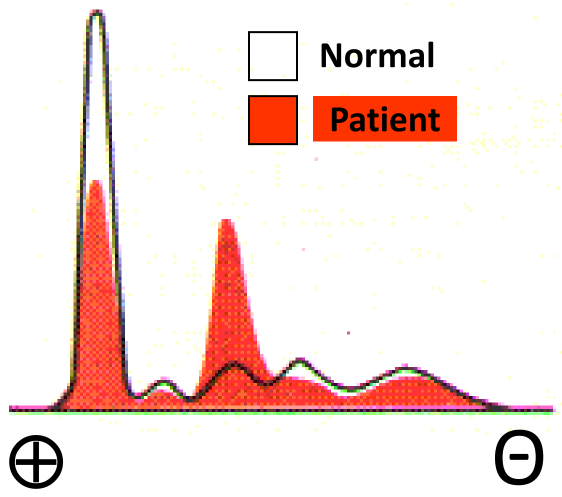


- Findings are from a patient with: _______
- Why is serum albumin low in this patient?

Case 3

A 4-year-old boy is brought to the physician because of a 2-week history of progressive swelling of her eyelids, face and feet. The face swelling is worse in the morning. Densitometry of serum protein electrophoresis is shown.

- Identify the bands that are abnormal in the patient
- Findings are from a patient with:________
- Which protein levels are most likely elevated?
- Which protein level is low in this patient?
- What are the clinical features of low levels of this protein?

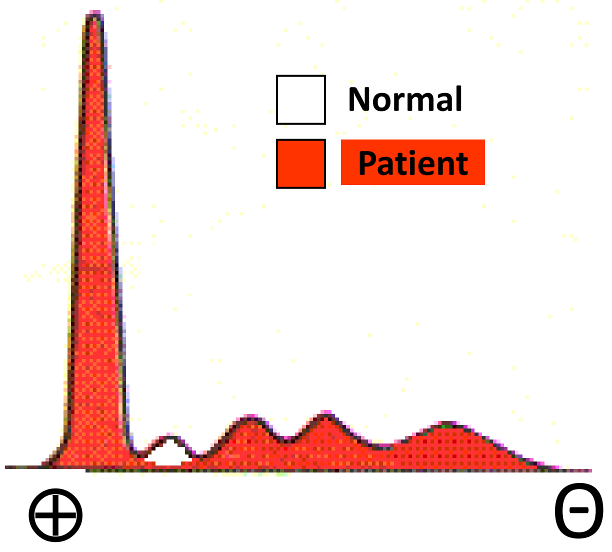


SOM.MK.I.BPM1.1.CPR.2.BCHM. 0326

Case 4

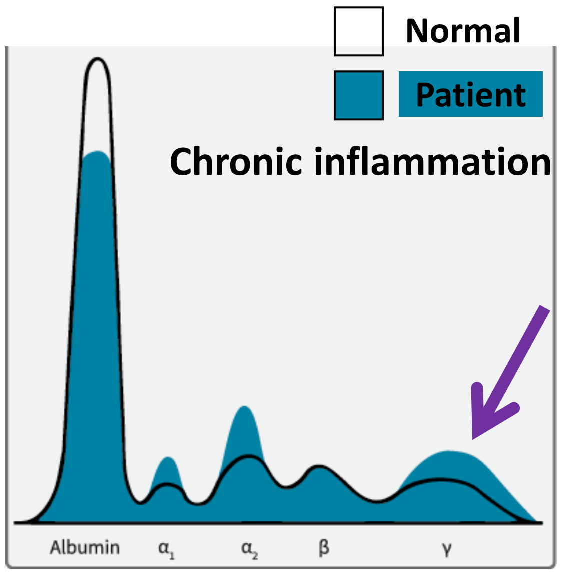
A 10-year-old boy is brought to the physician by his parents because of a 2-month history of yellowish discoloration of eyes and a loss of appetite. Physical examination shows a sharp border of the liver. Densitometry of serum protein electrophoresis is shown.

- Identify the bands that are abnormal in the patient
- Name the component proteins of this band
- Findings are from a patient with: ___________
- What is the normal function of the protein deficient in this patient?
- What are the organs affected in this disorder?
- Identify the mode of inheritance and explain allelic heterogeneity.

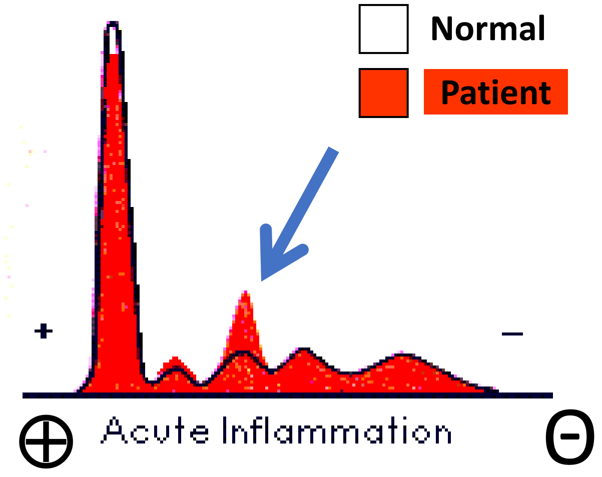


Case 5

Densitometry of serum protein electrophoresis of two patients is shown.



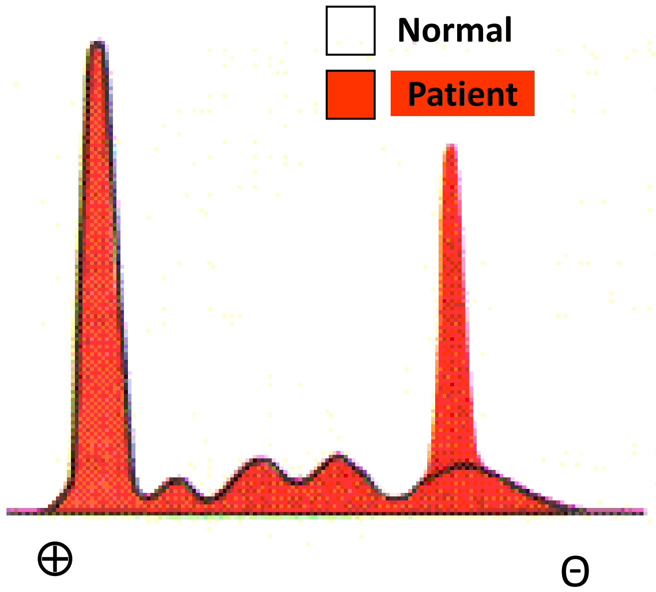
- Which protein bands are increased in acute and chronic inflammation?
- Which protein is decreased in the patients? Explain why.
- Name the component proteins in the band shown (blue arrow) and identify the function of each protein
- Which protein is measured to monitor progress of inflammation?
- Name proteins in the region shown by purple arrow. Which cells synthesize these proteins?



SOM.MK.I.BPM1.1.CPR.2.BCHM. 0326

Case 6

A 58-year-old man comes to the physician because of a 1-month history of low back pain. An x-ray of the back shows lytic lesions in the vertebrae and compression fractures of L2, L3 and L4. The x-ray of the left arm is shown. Densitometry of the serum protein electrophoresis is shown.



- Identify the bands that are abnormal in this patient
- Name the cells that produce these proteins

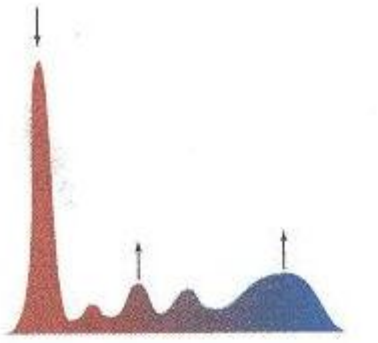
- Findings are from a patient with:__________
- What are the urine findings in this patient?

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0326

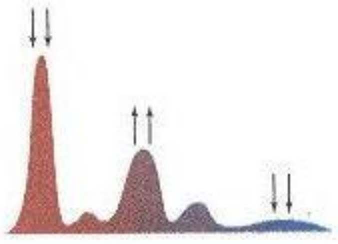
Summary:

Serum protein	Function	Clinical significance
Albumin	Maintains colloid osmotic pressure (oncotic pressure) Transport of ions (calcium), bilirubin, hormones, fatty acids	Hypoalbuminemia: Liver cirrhosis; Kwashiorkor; Nephrotic syndrome; Burns; Inflammation (Negative APR)
Alpha1-antitrypsin	Inactivates neutrophil elastase	Deficiency: COPD (emphysema); Liver cirrhosis (some patients)
Alpha-feto protein (AFP)	Transport protein in fetus	High: **Liver cancer;** Ovarian or testicular cancer Maternal AFP: High (Neural tube defect); Low (Down syndrome)
Alpha2-macroglobulin	Inactivates serum enzymes	High: Nephrotic syndrome
Haptoglobin	Binds free hemoglobin	Low: Hemolytic anemia; High: Inflammation (APR)
Ceruloplasmin	Binds copper	Low: Wilson disease (KF ring; liver; basal ganglia) High: Inflammation (APR)
Transferrin	Transport protein for iron	Low: Inflammation (Negative APR)
Hemopexin	Binds free heme	High: Inflammation (APR)
Beta-lipoprotein (LDL)	Transports cholesterol to tissues from liver	High: Familial hypercholesterolemia; Nephrotic syndrome
Gamma-globulins (immunoglobulins)	Provide immunity (synthesized by plasma cells; activated B-lymphocytes)	Low: Hypogammaglobulinemia High: Inflammation (Polyclonal); Multiple myeloma (monoclonal); Liver cirrhosis (beta-gamma bridge)

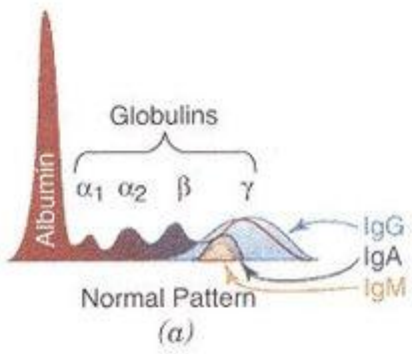
Serum protein electrophoresis patterns

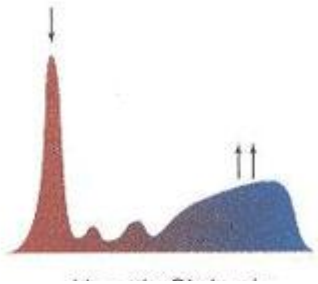


Chronic inflammation

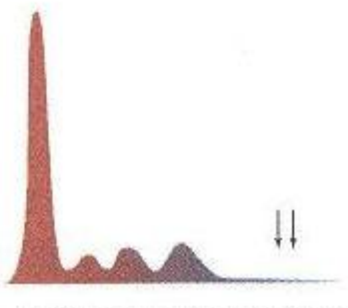


Nephrotic syndrome

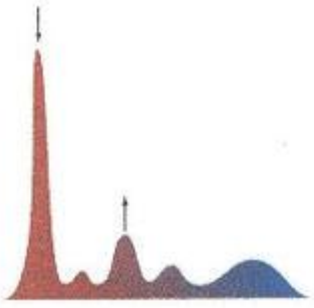




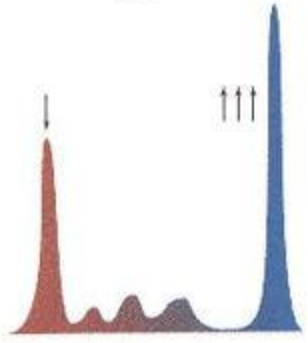
Liver cirrhosis



Hypogammaglobulinemia



Acute inflammation



Multiple myeloma